The Last Mile to First Treatment: Search under Uncertain Medication Availability

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Consumers search for goods whose availability they cannot verify in advance. When travel costs compound this uncertainty, purchases may fail despite demand. I study these frictions in the market for buprenorphine, the leading medication for opioid use disorder. In Washington administrative claims, fewer than half of first-time patients fill their prescriptions. Normalizing each patient's default pharmacy to zero search cost identifies search costs and preferences: a purchase elsewhere reveals further search was worthwhile. Search frictions explain 70 percent of treatment failures. Guiding patients to available pharmacies captures most of the gains from requiring every pharmacy to carry the drug.

Keywords: sequential search, medication initiation, pharmacy availability

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I. Introduction

Many consumer markets pose two problems for empirical work on search. First, researchers observe what consumers buy but not how they searched. Second, consumers may have to learn both the value of each option and whether the product is available. Final choices then reflect three underlying forces: availability, preferences, and search costs. I study these problems in the market for buprenorphine, the leading medication for opioid use disorder (OUD). Opioid overdoses now kill more than 50,000 Americans each year, making OUD one of the nation's deadliest public health crises.[1] Buprenorphine reduces withdrawal, cravings, relapse, and overdose risk (Larochelle et al., 2018; Ma et al., 2019; Dever et al., 2024), yet 87% of individuals with OUD remain untreated (Krawczyk et al., 2022). Even among patients who receive a prescription, more than half fail to fill it. Why does treatment break down in the last mile, after a prescription has been written?

Two features of the pharmacy market turn a prescription into a search problem. First, availability is limited: only 56% of pharmacies in Washington carried buprenorphine at least once during the study period. Second, availability is hard to verify. Pharmacies are not required to disclose inventory and often decline to confirm availability by phone, citing concerns about diversion and theft. Even addiction specialists lack reliable inventory information, leaving patients to identify a pharmacy on their own. These barriers were especially acute when controlled substances were prescribed on paper rather than transmitted electronically.[2] Patients often had to travel to pharmacies merely to learn whether buprenorphine was available. Search costs therefore reflect both physical travel and uncertainty about whether a trip will produce a fill. Policies such as real-time inventory information and first-fill incentives target different margins of this last-mile problem, making it important to know which margin prevents initiation.

I combine individual-level insurance data with a structural model of sequential search. My primary source is the Washington State All-Payer Claims Database (WA-APCD), which links medical and pharmacy claims across Medicaid, Medicare, and commercial insurers from 2014 to 2019. Medical claims identify patients with OUD, while pharmacy claims record whether and

[1] Opioid-involved overdose deaths, CDC WONDER Multiple Cause of Death database [Accessed: 2025-07-18].

[2] In 2018, only 15.9% of controlled substance prescriptions in Washington State were transmitted electronically, according to Surescripts: Surescripts 2018 National Progress Report [Accessed: 2025-07-18].

where patients fill buprenorphine prescriptions.[3] To my knowledge, this is the first study to use detailed individual-level insurance claims for patients with OUD, a population typically excluded from administrative claims because of federal privacy protections. This data lets me define each patient's default pharmacy from past purchase patterns and compare that default with the patient's eventual buprenorphine fill decision, variation that aggregated data cannot provide. To measure pharmacy availability, I link the claims to the Automated Reports and Consolidated Ordering System (ARCOS), which records buprenorphine shipments to every retail pharmacy in Washington.

Buprenorphine initiation shares two features that make search difficult to study in many markets: researchers observe final choices but not the search process, and consumers face uncertain product availability. I observe prescription fills but not the pharmacies patients considered or visited, while patients do not know which pharmacies carry the medication before they search. I develop a model of patients' search for buprenorphine that embeds the sequential-search insight of Moraga-González, Sándor and Wildenbeest (2023) in an environment where search behavior is not directly observed and availability is uncertain.

Low initiation can reflect weak preferences, high search costs, or both.[4] I discipline this distinction using a salient feature of prescription-filling behavior: most patients concentrate their fills at a default pharmacy, defined as the pharmacy they visit most often. In my sample, 93% of patients fill most of their prescriptions at a single pharmacy, and buprenorphine initiation falls from 65% to 15% when the default does not carry the drug. These patterns motivate the assumption that patients always search their default pharmacy, which is equivalent to normalizing its search cost to zero. A fill at a non-default pharmacy then implies that searching beyond the default was worthwhile. Under this normalization, default-pharmacy utility captures preferences, while non-default utility reflects preferences net of search costs, yielding a tractable

[3]The data also include demographics such as age, sex, insurance type (Medicaid versus other), and residential ZIP code, which I use to measure travel distances to nearby pharmacies. I supplement individual characteristics with ZIP Code Tabulation Area–level data from the American Community Survey.

[4]Two routes can separately identify preferences and search costs. Exclusion restrictions (a covariate that enters only one component) identify the slope coefficients but preclude the demographic overlap I want and leave the intercept unidentified, so I do not recover the magnitude of search costs, which my counterfactuals target. A second route adds moments that pin down search costs, such as the survey-based search counts in Moraga-González, Sándor and Wildenbeest (2023). My default-pharmacy normalization instead allows demographics to enter both components and recovers the magnitude of search costs from purchase decisions alone.

logit-based estimation problem.

I also extend the search model to searched options that may be unavailable. Standard search models assume that visiting a seller reveals match value for an option that can be purchased. Here, search also reveals whether the option can be purchased at all: a pharmacy that does not carry buprenorphine cannot generate a fill. I capture this feature by embedding availability beliefs into search costs. The resulting *effective search cost* rises as perceived availability falls; if a patient believes a pharmacy has only a 50% chance of carrying buprenorphine, the effective cost of visiting it doubles.

The estimated model shows that search frictions explain 70% of treatment failures. Traveling one additional mile raises the effective search cost by \$0.62, about 3% of the average out-of-pocket cost for buprenorphine (\$21.33). Fewer than 40% of patients search more than once, so many rely on a single pharmacy even when it does not carry buprenorphine. The burden is uneven: Black patients, Medicaid and Medicare enrollees, and patients with severe OUD are less likely to search beyond their default pharmacy and have lower estimated preferences for buprenorphine.

The counterfactuals compare policies that expand supply with policies that reveal where supply already exists. Under rational beliefs about local pharmacy availability, a universal availability mandate raises initiation from 45% to 58%, adding 13 initiates per 100 first-time OUD patients with a buprenorphine prescription. Real-time inventory information for prescribers raises initiation by 17%, adding 8 initiates per 100 patients and capturing most of the mandate's effect without requiring every pharmacy to stock the drug. A smaller intervention, prescriber relationships with five nearby pharmacies known to carry buprenorphine, raises initiation by 7%. Using back-of-the-envelope health and fiscal conversions, every 100 additional initiates avert about 60 emergency room visits per year, reduce annual hospital costs by \$219,180, and prevent about 13 overdoses, including 2 fatal overdoses, over five years.

E-prescribing, now mandatory nationwide, implements part of this information policy in practice. It lets pharmacies confirm availability before patients travel, but it can also raise switching costs because controlled-substance prescriptions generally require the prescriber to issue a new order when the selected pharmacy is out of stock. In the model, e-prescribing raises initiation on net unless switching costs more than double. Pairing e-prescribing with a small first-fill subsidy targets the remaining physical-search margin: calibrating the additional switching burden as a 20% increase in search costs, a $13.49 rebate for Medicaid patients (the most price-sensitive and least likely to initiate) raises initiation to 70%, comparable to rates for other chronic medications. With e-prescribing already in place, the counterfactual points to a moderate one-time initiation subsidy for Medicaid patients as a complementary tool for closing the last mile.

This paper contributes to the empirical literature on sequential search, beginning with Weitzman (1979) and recently advanced by Moraga-González, Sándor and Wildenbeest (2023), who use insights from Armstrong (2017) and Choi, Dai and Kim (2018) to recast dynamic search as a static discrete choice problem. First, I allow searched options to be unavailable: because a visited pharmacy may not carry buprenorphine, search reveals not only an option's match value but whether it can be purchased at all. This feature is relevant beyond pharmacies whenever products stock out (e.g., Conlon and Mortimer, 2013) or a natural disaster disrupts supply (e.g., Luco, Klopack and Lewis, 2024), leaving consumers to search for available substitutes. Second, I identify the model without the direct search data that most applications require (e.g., Kim, Albuquerque and Bronnenberg, 2010; Santos, Hortaçsu and Wildenbeest, 2012; Honka, 2014; Kim, Albuquerque and Bronnenberg, 2017; Ursu, 2018; Compiani et al., 2024). I recover the relevant objects from final choices alone by using a natural option, the patient's default pharmacy, as the zero-search-cost normalization: choosing a non-default option reveals at least one additional search and the cost it must have covered. Third, I connect this default-based approach to the literature on consideration sets and default options in consumer choice, particularly in healthcare (Hortaçsu, Madanizadeh and Puller, 2017; Abaluck and Adams-Prassl, 2021). Existing models typically treat consideration as binary, sorting consumers into those who consider only their default and those who consider the full choice set (Handel, 2013; Ho, Hogan and Morton, 2017; Heiss et al., 2021). Embedding the default in sequential search generates the intermediate cases between these extremes: consumers evaluate a limited set of options whose size depends on search costs, a parsimonious and behaviorally grounded alternative to the hybrid model of Abaluck and Adams-Prassl (2021).

This paper also contributes to the literature on policy responses to the opioid crisis. Many interventions restrict opioid supply, including the 2010 OxyContin reformulation (Severtson et al., 2013; Alpert, Powell and Pacula, 2018; Evans, Lieber and Power, 2018) and DEA crackdowns on rogue distributors (Soliman, 2023; Gui, 2024; Donahoe, 2025). These efforts curtailed access to prescription opioids but often failed to improve health outcomes and sometimes accelerated substitution into illicit markets. Demand-side policies have shown greater promise: expanding naloxone access reduces overdose mortality (Abouk, Pacula and Powell, 2019; Rees et al., 2017), and increasing the number of authorized buprenorphine prescribers improves treatment initiation (Gui, 2024). Using availability at a patient's default pharmacy as an instrument, I find that initiation reduces emergency room visits by 20% relative to similar patients, offering causal confirmation of earlier observational findings (Sullivan, Szczesniak and Wojcik, 2021; Skains et al., 2023; Yarborough et al., 2024). Reducing last-mile barriers can therefore amplify upstream policies that expand treatment access.

The remainder of the paper proceeds as follows. Section II describes the institutional setting. Sections III and IV present the data and key empirical patterns. Section V develops the sequential search model under uncertain availability and the estimation strategy. Sections VI and VII report the estimation results and counterfactual policy analyses. Section VIII concludes.

II. Institutional Background

This section establishes the institutional context for the two barriers central to this paper: limited pharmacy availability and patient uncertainty about where buprenorphine can be obtained.

A. Medication for Opioid Use Disorder

> "If you were to just imagine a medicine, or a chemical compound, that could stop the opioid epidemic, that medicine would probably look a lot like buprenorphine."
> —Ethan Brook, The Atlantic

Buprenorphine is one of three medications approved by the U.S. Food and Drug Administration for treating OUD, alongside methadone and naltrexone. Among these, buprenorphine is distinguished by its safety profile and flexibility in administration, making it the most commonly prescribed treatment in office-based settings.

Methadone, a full opioid agonist, is subject to tight federal restrictions. It can only be dispensed through licensed opioid treatment programs, and patients are generally required to attend daily visits during the first 90 days of care.[5] Naltrexone, an opioid antagonist, is more often prescribed for alcohol use disorder and requires complete detoxification prior to initiation, a barrier many patients are unable to overcome. In contrast, buprenorphine is a partial agonist with a ceiling effect that mitigates overdose risk while effectively suppressing cravings and withdrawal symptoms. Certified prescribers can offer it in office-based settings, expanding access beyond the confines of OTPs. During the study period, over 80% of pharmacologically treated OUD patients received buprenorphine.[6]

The treatment pathway for buprenorphine resembles that of other long-term therapies for chronic conditions. Patients begin with an induction phase focused on clinical stabilization, followed by maintenance therapy that may extend for months or years. Behavioral therapy and peer support are often encouraged as complementary elements. Addiction specialists frequently liken buprenorphine to "insulin for diabetes" (McLellan et al., 2000). It is not a cure, but a cornerstone of sustained disease management that enables individuals to return to functional daily life.

Initiating buprenorphine treatment is therefore a pivotal step in the recovery process. As with other chronic illnesses, early access and continuity of care are critical to achieving favorable outcomes (D'Onofrio et al., 2015; Wakeman et al., 2020). Yet frictions at the pharmacy level can obstruct treatment even for patients who have already obtained a prescription.

[5] Following regulatory changes during the COVID-19 pandemic, take-home methadone doses became more accessible. This flexibility has since been extended at the discretion of the prescribing practitioner. See SAMHSA guidance: Methadone Take-Home Flexibilities Extension [Accessed: 2025-07-19].

[6] Buprenorphine is available in two formulations: one combined with naloxone (commonly branded as Suboxone) and one without. The combination is recommended as first-line treatment due to its deterrent effects on misuse. The mono-product is typically reserved for patients contraindicated for naloxone, such as pregnant women. Nearly all prescriptions in this study pertain to the combination formulation.

B. Pharmacy Stocking Behaviors

Despite buprenorphine's central role in treating OUD, its availability at retail pharmacies remains limited. Nationally, only 57.9% of pharmacies stocked the medication as of 2022 (Weiner et al., 2023). In Washington (2014–2019), only 56% of pharmacies that received any opioid shipments ever received buprenorphine. Given that its therapeutic role is often compared to insulin for diabetes, the reluctance of pharmacies to carry, let alone consistently stock, buprenorphine is striking.

The reasons pharmacies choose not to carry buprenorphine are not fully understood. A policy roundtable convened by the Substance Abuse and Mental Health Services Administration identified several commonly cited barriers.[7] First, pharmacies may fear stigmatization, including concerns that stocking buprenorphine could attract clientele perceived as undesirable. Second, regulatory uncertainty, particularly regarding ordering thresholds, may discourage pharmacists from requesting inventory, out of fear of triggering compliance violations.[8] Third, the economics of dispensing buprenorphine are often unfavorable. Pharmacists report high labor costs associated with verifying prescriptions and navigating insurance coverage, paired with low reimbursement rates. The same roundtable reports that pharmacies lose approximately $10 per buprenorphine fill.[9]

The challenge extends beyond whether a pharmacy ever stocks buprenorphine. Among those that do, inventory is often inconsistent or chronically insufficient. Because buprenorphine must be initiated within a narrow window after the onset of withdrawal, even short delays can disrupt treatment. As a result, patients are frequently forced to search across multiple pharmacies, introducing frictions that delay or deter initiation altogether.

[7] See: SAMHSA Policy Priority Roundtable on Buprenorphine Access in Pharmacies [Accessed: 2025-07-19].

[8] The Drug Enforcement Administration shut down Oak Hill Hometown Pharmacy, citing an "imminent danger to public health and safety" due to its high volume of buprenorphine dispensing. Although a federal judge later ruled in the pharmacy's favor, noting it had acted professionally and within legal bounds, the DEA's action effectively ended its operations. Wholesale suppliers, wary of regulatory scrutiny, refused to resume shipments. See NPR Coverage [Accessed: 2025-07-19].

[9] A natural question is why pharmacies continue to carry buprenorphine despite the apparent financial loss. Although the buprenorphine fill itself may be unprofitable, pharmacies can recoup losses through sales of other medications, particularly generics, as patients often have multiple prescriptions. In addition, pharmacies may fear losing the patient entirely if they do not carry buprenorphine. This behavior is evident in the data: about 40% of first-time patients switch all of their prescription fills to a pharmacy that carries buprenorphine, leaving their previous default pharmacy.

C. Frictions in Learning About Availability

Patients cannot easily resolve uncertainty about buprenorphine availability before visiting a pharmacy. Pharmacists are not legally required to disclose inventory levels, and many adopt a policy of nondisclosure for controlled substances. This practice, especially common in the early 2010s amid heightened fears of diversion, reflects concerns that confirming stock over the phone could attract individuals seeking opioids for non-medical use or increase the risk of theft.

These concerns appear in professional discourse. In a 2014 thread on *studentdoctor.net*, a forum used by pharmacists, contributors described common responses to telephone inquiries:

- "If it's a control, no matter what I say: 'out of stock.' I don't want any more of 'those' types of patients."
- "Just ask for the strength and quantity. Pretend to check. Then politely say no."

Many pharmacies will not even check inventory without first seeing a valid prescription. During the study period, most prescriptions for controlled substances were handwritten and physically carried by the patient. As of 2018, only 15.9% of controlled substance prescriptions were transmitted electronically. This meant that patients often had to visit pharmacies in person simply to determine whether the medication was available.

Even if e-prescribing reduces uncertainty about inventory, it may raise the cost of switching pharmacies by requiring patients to contact their provider each time a redirection is needed. As a result, patients frequently initiate searches only after a failed fill attempt at their default pharmacy, and often without reliable information about which pharmacies are likely to carry buprenorphine.

III. Data

To study pharmacy search behavior and buprenorphine access in Washington State, I use three primary datasets. Table E1 reports summary statistics for the main variables used.

A. *Washington State All-Payer Claims Database*

The WA-APCD includes medical and pharmacy claims for most insured residents of Washington State, covering public payers (Medicaid and Medicare) as well as a subset of private plans. It provides detailed claim-level information on patient demographics, diagnoses, procedures, provider identifiers, payment amounts, and residential ZIP codes. Linked at the patient level, the WA-APCD enables reconstruction of individual medical histories, including prior diagnoses of substance use disorder and prior-quarter opioid consumption, and supports linkage between medical visits and subsequent pharmacy fills.

A unique feature of the WA-APCD is that it includes patients diagnosed with OUD, a subset of substance use disorders historically excluded from administrative claims data due to federal privacy protections. To my knowledge, this is the first paper to use detailed patient-level insurance claims to study treatment-seeking behavior among individuals with an OUD diagnosis.

I restrict the sample to first-time OUD diagnoses between 2014 and 2019. Using medical claims, I identify the initial OUD-related visit and track whether the patient subsequently filled a buprenorphine prescription and at which pharmacy. The analysis further focuses on patients seen by providers who routinely prescribe buprenorphine, making the index visit a high-intent treatment opportunity; episodes in which patients are referred to methadone or other alternative treatment pathways are excluded where observable. The relevant empirical object is therefore not all OUD diagnoses, but initiation following encounters with providers whose practice patterns indicate buprenorphine treatment intent. Further details on sample construction are in Appendix A.A1.[10]

B. *Automated Reports and Consolidated Ordering System*

ARCOS is a national surveillance system maintained by the Drug Enforcement Administration under the Controlled Substances Act of 1971. It tracks opioid shipments across the supply chain, from manufacturers and distributors to hospitals and pharmacies, recording the shipment

[10]The WA-APCD data I have access to cover an OUD-adjacent subpopulation: patients with an OUD diagnosis, patients who received buprenorphine, and patients dispensed opioids four or more times in any calendar year. This last group is included because frequent opioid use is strongly predictive of OUD.

date, quantity, dosage, National Drug Code, and the identities of sender and recipient for each transaction.

ARCOS enables the construction of daily pharmacy-level buprenorphine inventory. Pharmacy claims record drug formulation, strength, and fill date, which I use to measure daily outflows; ARCOS shipments provide the corresponding inflows. Subtracting cumulative claims from cumulative shipments, I estimate whether a pharmacy had buprenorphine available on a given day, including the date of a patient's prescription. This measure has three limitations. First, not all buprenorphine is dispensed through insurance claims, so some outflows are unobserved. Second, I treat each unit at its labeled dosage and cannot account for partial dispensing: a 16mg tablet cannot be split to fill two 8mg prescriptions. Third, I abstract from formulation differences: a pharmacy stocking buprenorphine tablets cannot substitute for a patient prescribed the sublingual film, even if the active ingredient is identical. All three factors cause me to overestimate pharmacy availability.[11] Construction details appear in Appendix A.A2.

C. Other Data Sources

ZCTA-Level Demographics. Individual-level demographic data in the WA-APCD are limited, so I supplement the analysis with ZIP Code Tabulation Area-level characteristics from the American Community Survey 5-year estimates, including median income, educational attainment (above high school), poverty rate, and unemployment rate.

Provider Characteristics. I use the National Plan and Provider Enumeration System (NPPES) to obtain information on prescribers, including specialty, gender, and years of experience. Historical NPPES snapshots were accessed through the NBER data repository.[12]

Product Classification. To distinguish buprenorphine formulations intended for OUD from those used for pain management, I scrape product descriptions and labeled indications from the DailyMed website by querying National Drug Codes for buprenorphine products.

[11] Overestimating availability biases against finding large search frictions: cases where the default pharmacy was truly unavailable are misclassified as available, attenuating the measured initiation drop when the default lacks buprenorphine. Estimated search costs are therefore a lower bound on the true frictions patients face.

[12] Link to NBER collection [Accessed: 2025-01-15].

IV. Stylized Facts

Three empirical patterns motivate the analysis. First, most patients consistently fill prescriptions at a single pharmacy, making default pharmacy behavior the norm. Second, buprenorphine initiation is low, with much of the shortfall driven by unavailability at patients' default pharmacies. Third, initiation is a decisive margin: patients who fill their first prescription are far more likely to remain in treatment, while those who do not rarely initiate later. I benchmark buprenorphine against levothyroxine, the standard treatment for hypothyroidism.[13] Levothyroxine is widely prescribed, routinely available, and not subject to the restrictions governing controlled substances. Because both patient groups come from the same population, differences in fill rates reflect pharmacy availability and search frictions rather than patient composition.

Appendix B presents three additional facts. First, buprenorphine prescriptions are typically filled within a narrow therapeutic window, underscoring the importance of timely search. Second, when the default pharmacy lacks buprenorphine, patients substitute to pharmacies near their home ZIP code or near the default pharmacy, consistent with sequential search. Third, using default availability as an instrument, I estimate that initiation reduces subsequent emergency room visits by at least 20%.

A. Default Pharmacy Behavior

I define an individual's default pharmacy as the pharmacy at which they fill most of their prescriptions prior to diagnosis.[14] Figure 1 shows that default pharmacy behavior is the norm for both groups: 93% of buprenorphine patients and 98% of levothyroxine patients filled most of their prior prescriptions at a single pharmacy.

Default pharmacy behavior reflects the architecture of e-prescribing. Because most patients' prior prescriptions were for non-controlled medications, their fills were routed electronically to a preferred pharmacy. E-prescribing platforms reinforce this default by auto-filling the last-used location, and medical guidelines encourage consolidating fills at a single pharmacy to reduce

[13]Hypothyroidism affects roughly 5% of the U.S. population.

[14]In rare cases, individuals may split prescription volume evenly across multiple pharmacies. If so, I define the default pharmacy as the one nearest to the individual's home ZIP code. In addition, if an individual moves during the year, as indicated by a change in residential ZIP code, I assign the default based on the pharmacy most frequently visited prior to the move.

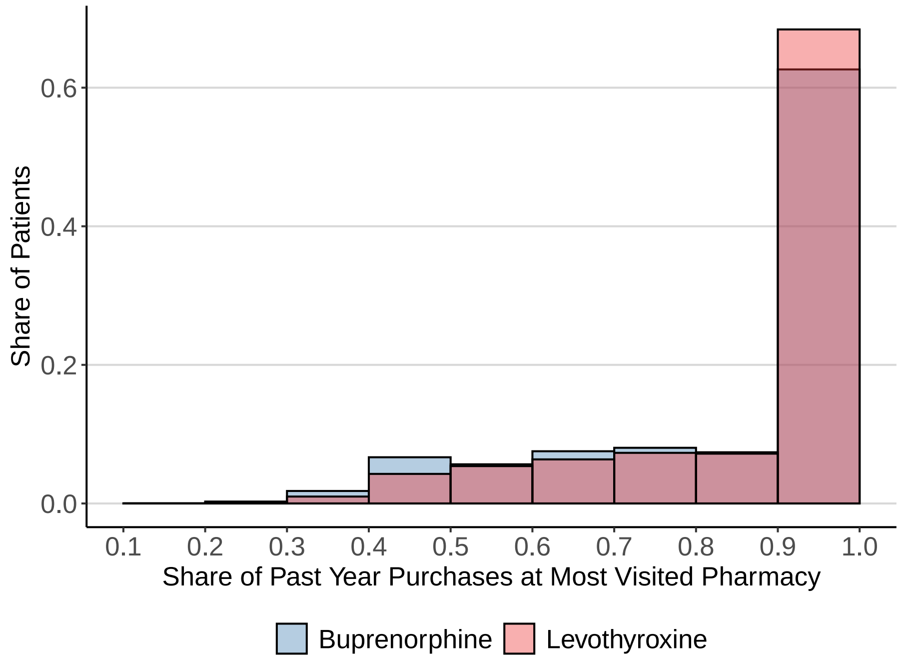


Figure 1. Concentration of Prior Fills at the Default Pharmacy

Note: Distribution of prior-fill concentration at the default pharmacy for first-time buprenorphine and levothyroxine patients. The default pharmacy is the pharmacy at which a patient fills most of their prescriptions in the year before diagnosis. The x-axis shows the share of prior fills at the default pharmacy; the y-axis shows the share of patients.

adverse drug interactions (Marcum et al., 2014). This concentration motivates a key modeling choice: I normalize search costs at the default pharmacy to zero and identify costs at non-default pharmacies from deviations when the default lacks buprenorphine.

Default pharmacies are not randomly assigned, raising an endogeneity concern for treating the default as the zero-search-cost first stop. Patients whose default pharmacy carries buprenorphine may differ systematically from those whose default does not. Stocked defaults may be located in better-served neighborhoods or serve patients with greater latent willingness to initiate treatment. If so, the relationship between default availability and initiation would partly capture unobserved patient and market characteristics rather than only the reduction in search costs, and the model would overstate the role of search frictions.

Two tests address this concern. First, I run a placebo test using levothyroxine refills, which should not respond to buprenorphine inventory at the default pharmacy. Conditional on year and pharmacy fixed effects, default buprenorphine availability does not predict levothyroxine refills either before or after the index OUD visit (Table E2), so stocked defaults are not simply proxying for a general propensity to fill prescriptions. Second, I exploit high-frequency within-pharmacy variation in availability. The identifying variation does not come from cross-sectional differences between pharmacies that do and do not carry buprenorphine; it comes from whether the same default pharmacy is stocked or temporarily unstocked on the prescription date. Because the default is fixed by prior prescription history, default-pharmacy fixed effects difference out time-invariant selection across pharmacies. Conditional on these fixed effects, month-year fixed effects, ZIP-code controls, provider characteristics, and patient observables, the remaining variation reflects temporary inventory shocks driven by shipment timing and depletion, which is plausibly orthogonal to a patient's latent willingness to initiate treatment. The initiation gap remains substantial under this specification (Table E3).

B. Low Initiation in OUD Treatment

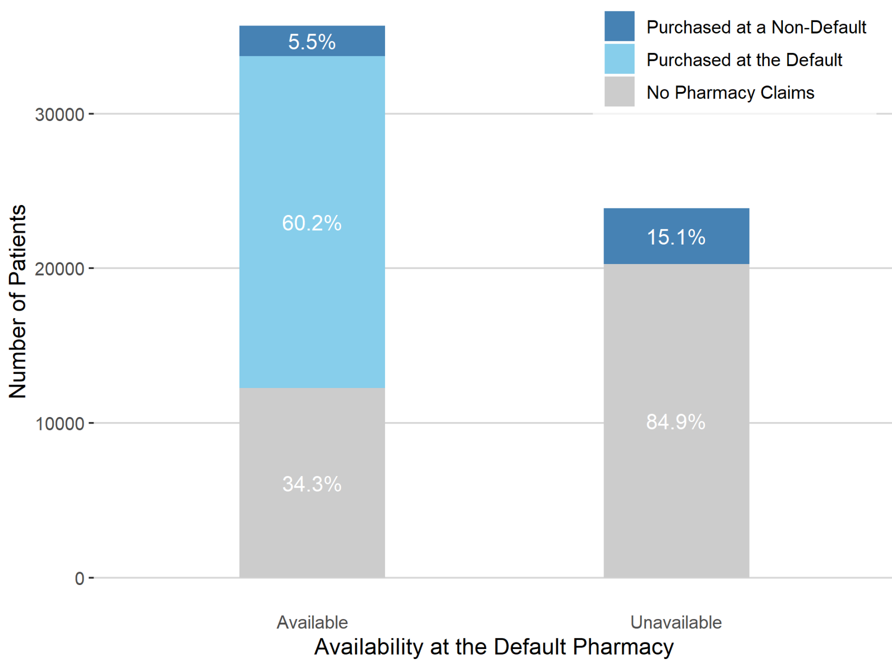
I classify patient behavior into three categories: filling at the default pharmacy, filling at a non-default pharmacy, or not filling at all.[15]

Figure 2a shows fulfillment shares conditional on buprenorphine availability at the default pharmacy on the prescription date. I approximate daily inventory by linking pharmacy claims to ARCOS shipment data. Three patterns emerge. First, although most patients had a default pharmacy with buprenorphine available, the majority of non-fills occurred when the drug was unavailable there. Second, availability at the default is associated with substantially higher initiation: only 34% of patients failed to fill when buprenorphine was available at their default. Third, when the default lacked the medication, just 15% of patients filled elsewhere, suggesting that most do not search extensively or cannot locate an alternative.[16]

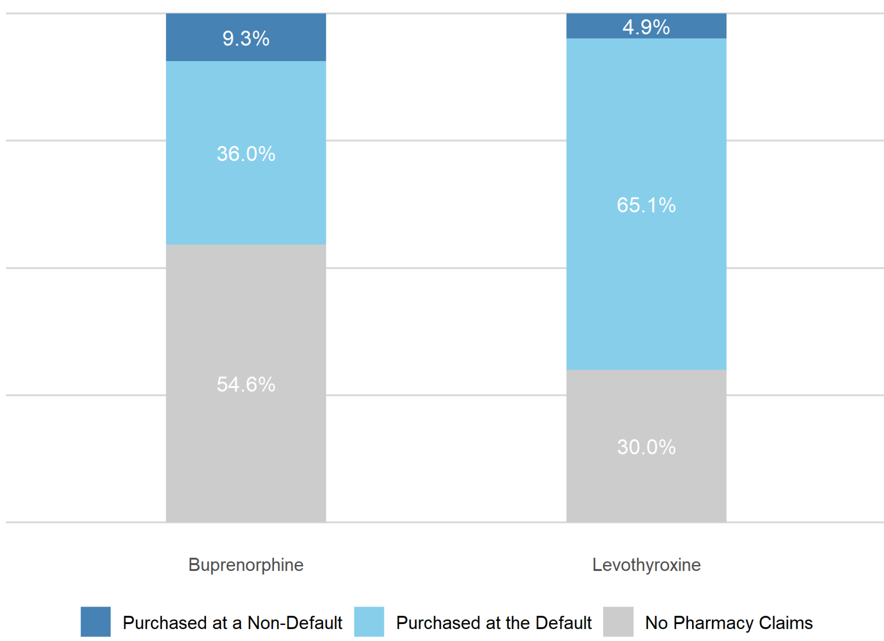
Figure 2b shows unconditional purchase patterns for first-time buprenorphine and levothyroxine patients. About 30% of hypothyroidism patients fail to fill following diagnosis, compared to

[15]This classification relies on medical claims data, which do not record the specific drug prescribed. Both hypothyroidism and OUD are life-threatening without medication, making prescribing highly likely. Among patients with a follow-up visit to the same provider, only 3% generated no pharmacy claims, further supporting this inference.

[16]For this group, I examine subsequent purchase patterns. As shown in Appendix B, patients tend to purchase from pharmacies near their home ZIP code or their default pharmacy, a pattern that motivates the structure of the search model.



(a) Purchase Location Conditional on Default Pharmacy Availability



(b) Purchase Location Following Initial Diagnosis

Figure 2. Pharmacy Purchase Patterns for New Patients

Note: Purchase location for first-time patients: default pharmacy, non-default pharmacy, or no fill. Panel (a) conditions on buprenorphine availability at the default pharmacy, defined as sufficient inventory on the prescription date. Panel (b) shows unconditional purchase shares for buprenorphine (OUD) and levothyroxine (hypothyroidism) patients.

nearly 55% of OUD patients. Among levothyroxine users, fills occur at the default pharmacy 13 times more often than at non-default locations; for buprenorphine, that ratio falls to 4:1, indicating buprenorphine patients likely need to search more than levothyroxine patients. When restricting to buprenorphine patients whose default has the drug available, fill rates and default usage closely resemble those of levothyroxine patients, suggesting that low initiation is not driven by patient non-adherence but by limited availability and the resulting search frictions. When the default lacks buprenorphine, 85% of patients fill nowhere else. This near-complete failure to search beyond the default is direct evidence of large search costs, and motivates modeling pharmacy choice as a sequential search problem.

C. Initiation Shapes Future Use

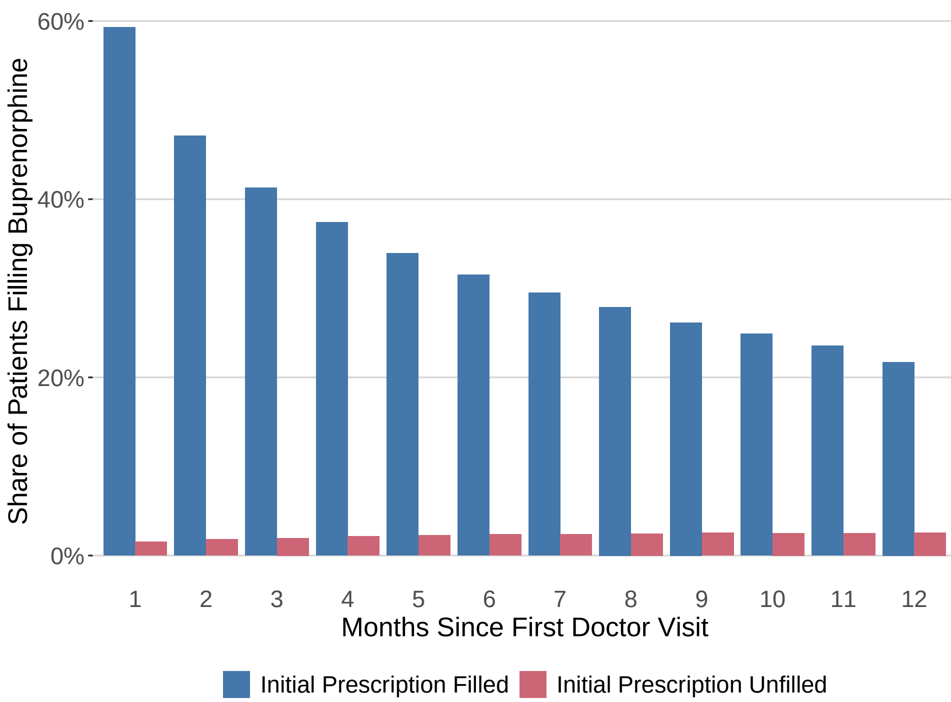


Figure 3. Monthly Buprenorphine Fills Following Initial Diagnosis

Note: Monthly share of patients filling at least one buprenorphine prescription in the 12 months after their first OUD-related doctor visit, shown separately for patients who filled their initial prescription and those who did not.

Search frictions and pharmacy availability are most salient at initiation, before patients have established a pharmacy relationship; refills can often be coordinated in advance even when a pharmacy is temporarily out of stock. The first filled prescription is also a decisive margin: patients who begin treatment are far more likely to remain engaged, while those who do not initiate rarely start later.

Figure 3 shows the persistence of this gap. Among patients who filled their initial prescription, nearly 60% remained on buprenorphine the following month, and more than 20% continued one year later. Fewer than 3% of those who did not fill ever initiated within the following year. Every subsequent month shows markedly higher treatment rates among initiators, and failing to initiate effectively forecloses future treatment. I therefore model the initial prescription decision as the relevant margin: it is where search frictions operate most forcefully and where their consequences are most durable.

V. Model

I model first-time buprenorphine patients as sequentially searching across pharmacies under uncertainty about drug availability. The framework extends Weitzman (1979)'s sequential search model to allow encounters with unavailable pharmacies: visiting such a pharmacy yields zero reward, as the patient learns it does not carry buprenorphine only upon arrival.[17] I adopt Moraga-González, Sándor and Wildenbeest (2023)'s reformulation of the problem (building on Armstrong 2017 and Choi, Dai and Kim 2018): the consumer selects the option that maximizes the minimum of the reservation value and realized utility, so that estimation targets the eventual purchase decision, which is observed, rather than the search process, which is not. Other sequential search estimators require data on search sequences; Moraga-González, Sándor and Wildenbeest (2023) identifies the model by combining purchase moments with survey-based search counts that record how many pharmacies consumers visit. This paper extends that approach by constructing search information from the purchase outcome itself: the normalization at the default pharmacy means that a purchase at the default reveals no additional search, while a purchase at any non-default pharmacy reveals that the patient searched at least once beyond

[17]This is discussed in the Application section of Weitzman (1979), where box i yields either zero reward ("failure") with probability $1 - p_i$ or a positive reward ("success") with probability p_i.

the default. This variation, embedded in the purchase outcome, is sufficient to separately identify search costs and preferences without auxiliary survey data. The subsections develop the utility specification, search environment, discrete choice reformulation, and estimation strategy in turn.

A. *Utility Specification*

I consider a market with J pharmacies, indexed by $j = 1, 2, \ldots, J$. The choice set includes all pharmacies within 25 miles for patients in urban ZIP codes and within 50 miles for patients in rural ZIP codes. The indirect utility that consumer i derives from visiting pharmacy j and obtaining buprenorphine is:

$$u_{ij} = \beta_0 + \alpha_{ins} P_{ij} + X_{ij}\beta + \epsilon_{ij}, \tag{1}$$

where P_{ij} is the price paid by consumer i at pharmacy j, and the price coefficient α_{ins} varies by insurance type (Medicaid, Medicare, or commercial coverage).[18] The vector X_{ij} includes consumer, provider, and pharmacy characteristics: demographics (e.g., age, gender, race), health indicators (e.g., past-quarter opioid consumption, prior diagnoses of substance use disorder or mental illness), provider characteristics (e.g., addiction specialization, years of experience), and pharmacy characteristics (e.g., chain affiliation, same-chain indicator with the default pharmacy). The idiosyncratic term ϵ_{ij} is i.i.d. across pharmacies and follows a Type I Extreme Value distribution. It captures match-specific value between the consumer and pharmacy, revealed only upon visiting: the quality of pharmacist interactions, wait times, or perceived stigma. These factors are especially salient in OUD treatment, where a dismissive or hostile pharmacist may cause a patient to leave without filling the prescription. The outside option (not filling the prescription) has utility $u_{i0} = \epsilon_{i0}$.

Equation (1) applies to pharmacies where buprenorphine is available. For unavailable pharmacies, I set the realized utility to $u_{ij} = u_{i0} - \iota$, where ι is an arbitrarily small positive value,

[18] Prices in this setting are largely determined by insurance contracts and network agreements, limiting pharmacies' ability to adjust prices in response to consumer search. In contrast to standard search markets where firms may exploit inattention through markups (Moraga-González, Sándor and Wildenbeest, 2023), price endogeneity is not a primary concern here.

ensuring that unavailable pharmacies are strictly dominated by the outside option and thus never chosen.

Consumers have correct conjectures about equilibrium prices and observe X_{ij}, but must visit a pharmacy to learn ϵ_{ij} and actual prices, which can deviate from equilibrium.[19] I model any trip beyond the default as a discrete search attempt, originating from either the patient's home ZIP code or the default pharmacy.[20] Search is sequential with costless recall. After each visit, the consumer decides whether to purchase, continue searching, or opt out. Before searching, the consumer knows each pharmacy's location, the outside option utility, and the distribution F from which ϵ_{ij} is drawn.

Let $\tilde{c}_{ij}$ denote the effective search cost for patient i when visiting pharmacy j, reflecting both the physical cost of search and the patient's belief about availability. Effective search costs vary across consumers and pharmacies. I assume the cumulative distribution $F^{\tilde{c}}_{ij}$ depends on search cost covariates through a location parameter μ_{ij}. The distribution has full support, though only its non-negative portion affects behavior. The belief ϕ_{ij} represents patient i's subjective belief about availability at pharmacy j and is not directly estimated; it is absorbed into the effective search cost $\tilde{c}_{ij}$, so what is recovered from the data is the combined object, not the two components separately.

B. *Optimal Sequential Search*

I characterize optimal search using a modified Weitzman rule. Let $\phi_{ij} \in (0, 1]$ denote patient i's belief that pharmacy j has buprenorphine available.[21] A consumer searches pharmacy j when the expected gain exceeds the search cost; the expected gain depends on both the probability of availability and the utility gain conditional on availability.

Let r denote the highest utility realized so far. The expected gain from searching pharmacy

[19] This follows most of the consumer search literature (Diamond, 1971; Wolinsky, 1986; Anderson and Renault, 1999): consumers hold correct conjectures about equilibrium prices but may find a deviation price upon visiting. In principle, this gives pharmacies an incentive to deviate, since consumers do not observe actual prices before arriving. In this setting, however, pharmacies do not set their own prices in the short run, so this assumption does not affect the counterfactuals.

[20] I remain agnostic about whether the search originates from home or from the default pharmacy; both distances are included as covariates in the search cost function.

[21] The fact that $\phi_{ij} \neq 0$ arises mechanically from the model's construction. Interpreting $\phi_{ij} = 0$ as full certainty that pharmacy j does not carry buprenorphine implies that the patient would never search that pharmacy. Consequently, the estimation remains unchanged even when $\phi_{ij} = 0$ for a subset of pharmacies.

j is:

$$\phi_{ij} \underbrace{\int_r^\infty (z - r) dF_{ij}(z)}_{\equiv H_{ij}(r)}, \tag{2}$$

where $F_{ij}(z)$ is the cumulative distribution function of the realized utility when pharmacy j has buprenorphine available.

The expected gain has two components: the probability of availability ϕ_{ij} and the expected utility gain conditional on availability $H_{ij}(r)$. By assumption, an unavailable pharmacy yields realized utility strictly below the outside option, so its expected contribution is zero. The total expected gain is therefore $\phi_{ij} H_{ij}(r)$, with the unavailability case contributing zero.

The consumer searches pharmacy j when the expected gain exceeds the cost c_{ij}. The *reservation value* r_{ij} is the utility threshold at which the consumer is indifferent, defined by:

$$\phi_{ij} H_{ij}(r_{ij}) - c_{ij} = 0. \tag{3}$$

I define the *effective search cost* as:

$$\tilde{c}_{ij} \equiv \frac{c_{ij}}{\phi_{ij}}, \tag{4}$$

Here c_{ij} is the physical or cognitive cost of visiting pharmacy j, independent of beliefs. Equation (4) shows that lower expected availability inflates the effective search cost: if $\phi_{ij} = 0.5$, the effective cost doubles. This generalizes earlier models, which implicitly assume $\phi_{ij} = 1$ and estimate c_{ij} directly.

I estimate $\tilde{c}_{ij}$ as a function of distance from home or from the default pharmacy, pharmacy characteristics (such as chain affiliation), and patient demographics. This approach imposes no restriction on the correlation between physical costs c_{ij} and availability beliefs ϕ_{ij}, including in the unobservable components. The limitation is that the two are not separately identified: a patient may avoid a chain pharmacy such as CVS because verification is time-consuming (high c_{ij}) or because they expect it is unlikely to carry buprenorphine (low ϕ_{ij}). Their combined effect on behavior is fully captured by $\tilde{c}_{ij}$.[22]

This bundling carries a substantive implication: the finding that effective search costs account for 70% of treatment failures reflects the joint effect of physical search costs and availability uncertainty, and I cannot separately quantify how much is attributable to each. A patient who does not search beyond the default may face high travel costs, believe that other pharmacies are unlikely to carry buprenorphine, or both. The counterfactuals in Section VII are designed with this limitation in mind, targeting interventions that reduce the combined object rather than one component in isolation.

Since H_{ij} is decreasing and strictly convex, Equation (3) has a unique solution, $r_{ij} = H_{ij}^{-1}(\tilde{c}_{ij})$. Following Lemma 1 from Moraga-González, Sándor and Wildenbeest (2023), the reservation value is:

$$r_{ij} = \delta_{ij} + H_0^{-1}(\tilde{c}_{ij}) \tag{5}$$

where $H_0(r) = \int_r^{\infty}(z - r)dF(z)$, and δ_{ij} is the deterministic component of utility from Equation (1). The reservation value rises with match quality δ_{ij} and falls with effective search cost $\tilde{c}_{ij}$: a more attractive pharmacy is worth searching sooner; a costlier one requires a higher expected payoff to justify a visit.

C. Discrete Choice Problem

Armstrong (2017) and Choi, Dai and Kim (2018) show that optimal sequential search is equivalent to a max-min decision rule: the consumer selects the option that maximizes the minimum of the reservation value and the realized utility. I extend this result to incorporate availability uncertainty. Consumers visit pharmacies in descending order of reservation values, observing at each visited pharmacy whether buprenorphine is available and, if so, drawing the realized utility. An unavailable pharmacy yields utility strictly below the outside option. Search

[22] One alternative is to impose a scalar structure on ϕ_{ij} and directly assume the parametric distribution for c_{ij}, which implicitly requires that the unobservables in c_{ij} and ϕ_{ij} are uncorrelated. This restriction is difficult to justify. Another possibility would be to use settings with variation in beliefs, such as comparing handwritten prescription periods with e-prescription periods. Under e-prescriptions, patients learn availability before visiting, which would aid separate identification of ϕ_{ij}. However, such data are not available in this setting.

stops when the realized utility at a visited pharmacy exceeds the reservation value of the next-best unvisited option. A formal proof appears in Appendix C.

Pharmacies without buprenorphine are excluded from the choice set: patients cannot purchase from them, and their realized utility falls strictly below the outside option. The discrete choice problem is therefore defined over A_i, the set of pharmacies with buprenorphine available to consumer i. For each $j \in A_i$, the latent utility w_{ij} is the minimum of the reservation value r_{ij} and the realized utility u_{ij}:

$$w_{ij} = \min\{r_{ij}, u_{ij}\}. \tag{6}$$

I adopt the following parametric form for the effective search cost distribution, which yields a tractable closed form for the distribution of w_{ij}:

$$F^{\tilde{c}}(\tilde{c}) = \frac{1 - \exp\left(-\exp\left(-H_0^{-1}(\tilde{c}) - \mu_{ij}\right)\right)}{1 - \exp\left(-\exp\left(-H_0^{-1}(\tilde{c})\right)\right)}, \tag{7}$$

where μ_{ij} is the location parameter for the search cost distribution.

Under this specification, the induced distribution of w_{ij} follows a Type I Extreme Value distribution with location parameter $\delta_{ij} - \mu_{ij}$:

$$F^{w}_{ij}(w) = \exp\bigl(-\exp\bigl(-(w - (\delta_{ij} - \mu_{ij}))\bigr)\bigr). \tag{8}$$

This closed-form result relies on the particular choice of the effective search cost distribution and is formally derived in Proposition 1 of Moraga-González, Sándor and Wildenbeest (2023).

The latent utility of pharmacy j for patient i is then:

$$w_{ij} = \delta_{ij} - (1 - D_{ij})\mu_{ij} + \xi_{ij}, \tag{9}$$

$$w_{i0} = \xi_{i0}, \tag{10}$$

where D_{ij} is an indicator equal to one if pharmacy j is patient i's default,[23] δ_{ij} is the deterministic

[23]Each patient has exactly one default pharmacy, such that $\sum_j D_{ij} = 1$. While the framework could be extended to

utility component from Equation (1), and $\mu_{ij} > 0$ is the search cost location parameter, required for $F^{\tilde{c}}(\tilde{c})$ to be a proper distribution. I parameterize μ_{ij} using a log-exp functional form:

$$\mu_{ij} = \log\bigl(1 + \exp(\lambda_0 + Z_{ij}\lambda)\bigr), \tag{11}$$

where Z_{ij} includes observed search cost shifters that vary by consumer and/or pharmacy. These include distance from the patient's home ZIP code, patient demographics, and pharmacy chain affiliations.

The conditional choice probability that patient i selects pharmacy j is:

$$s_{ij} = \frac{\exp(\kappa_{ij})}{1 + \sum_{j \in A_t} \exp(\kappa_{ij})}, \tag{12}$$

where $\kappa_{ij} = \delta_{ij} - (1 - D_{ij})\mu_{ij}$. Pharmacies without buprenorphine are never chosen and are excluded from estimation.

This discrete choice framework yields a closed-form logit despite the underlying sequential search structure. Availability constrains the choice set, and search costs enter the utility index via μ_{ij}. Estimation requires no auxiliary data on search sequences or separately identified preference moments.

D. Estimation and Identification

I estimate the model via maximum likelihood. The log-likelihood function is:

$$\mathcal{L}(\theta) = \sum_{i=1}^{N} \sum_{j \in \{A_{it}, 0\}} Y_{ij} \log(s_{ij}), \tag{13}$$

where $Y_{ij} = 1$ if patient i fills their buprenorphine prescription at pharmacy j, and $Y_{ij} = 0$ otherwise. The parameter vector θ includes the price coefficient α, the preference parameters β, and the search cost parameters λ.

To reduce computational burden, I restrict each consumer's choice set to pharmacies within

allow multiple defaults or no default at all, I restrict attention to a single default pharmacy in order to define distance from the default in a consistent way.

25 miles for urban ZIP codes and 50 miles for rural ZIP codes. These radii are generous relative to observed search behavior: nearly all pharmacy visits in the data occur within these bounds. The restriction also removes a small number of implausible observations where recorded ZIP codes suggest patients may have moved without updating their records.

Identification. Identification rests on separating utility δ_{ij} from search costs μ_{ij} through two sources of variation. The normalization at the default is supported by the stylized facts in Section IV: 93% of buprenorphine patients filled most of their prior prescriptions at a single pharmacy, indicating that patients reliably return to the default without deliberate search. The default defines the patient's baseline access channel, and the estimated search-cost parameters capture the incremental friction of moving beyond it. Any fixed cost of reaching the default, including stigma, wait time, or the effort of presenting a prescription, is absorbed into the outside option and the utility of initiating treatment.

First, I exploit the asymmetry between default and non-default options. The default is always searched at zero cost, so its latent utility depends only on δ_{ij}. Non-default utilities reflect both δ_{ij} and μ_{ij}. Comparing default choices to the outside option (normalized to zero) identifies δ_{ij}; given that the combined effect $\delta_{ij} - \mu_{ij}$ is identified through non-default choices as in a standard logit, μ_{ij} follows by differencing.

As an illustration, let β_0 be the constant in δ_{ij} and λ_0 the corresponding constant in μ_{ij}. A change in β_0 shifts the default's utility but leaves the outside option unchanged, so comparing the two identifies β_0. Given the total effect of the constant is identified, λ_0 follows by differencing. This logic extends to any covariate entering both specifications.

Second, covariates that enter only one component are identified without additional assumptions. In my specification, the distance between the patient's home and the pharmacy enters only μ_{ij}, not δ_{ij}, and so directly identifies the search cost coefficient.

Although μ_{ij} follows the log-exp form in Equation (11), this imposes minimal constraint in practice. For moderate values, $\log(1 + \exp(\lambda_0)) \approx \lambda_0$: for example, $\log(1 + \exp(3)) = 3.05$, and the average estimated $\hat{\mu}_{ij}$ in my sample is 7.31. Functional form is not the primary source of identification.

Identification does not require closed-form reservation values. Any purchase at a non-default pharmacy reveals search: the buyer found the non-default's utility, net of search cost, to exceed both the default and the outside option. Because the default is always searched, its realization is known (as is the outside option), while non-defaults are evaluated only after incurring additional search cost. This asymmetry between the known default realization and non-default utility net of search provides the variation needed to recover search costs.

Correctly specifying the default is essential. I define it as the pharmacy where the patient most frequently filled prescriptions prior to the index visit. For non-controlled substances, providers typically transmit prescriptions to the patient's designated pharmacy on file, so a consistent fill history strongly indicates intentional selection. Misclassifying the default (for example, treating the nearest pharmacy as the default) would overstate search intensity and understate true search costs, since patients are more likely to fill at their true default than at the nearest pharmacy, conditional on availability.

VI. Results

I present the estimated coefficients and discuss their interpretation, then characterize predicted search behavior implied by the model. Appendix D discusses internal validity.

A. Estimated Coefficients

Table 1 reports estimated coefficients from the preference and search components of the model. A higher preference parameter $\hat{\delta}$ reflects greater utility from filling a prescription at a given pharmacy; a higher search cost parameter $\hat{\mu}$ indicates greater barriers to reaching that pharmacy.

Price and Travel Distance: The price coefficient is small in magnitude, indicating limited price sensitivity in the general population. Medicaid recipients are substantially more price-sensitive, consistent with lower incomes in this population. Search costs increase with travel distance from both the patient's residence and their default pharmacy, with a larger increment per mile in urban areas. Each additional mile of travel imposes an effective search cost of approximately \$0.62 from the patient's residence and \$0.32 from the default pharmacy, representing 2.9% and 1.5% of the average out-of-pocket cost for buprenorphine (\$21.33), respectively.[24] For

[24]Estimates are scaled by the share of patients on Medicaid and Medicare, and distances are population-weighted based

comparison, Moraga-González, Sándor and Wildenbeest (2023) estimate that traveling one kilometer in the European car market imposes an average cost of €148, or roughly 0.7% of the average vehicle price of €19,917. Travel costs thus represent a substantially larger share of product price in this market than in the automobile market.

Provider Characteristics: Provider characteristics entering the utility function should not be interpreted as reflecting prescribing decisions; they capture how provider attributes shape the perceived value of treatment for the patient. Patients assigned to female providers exhibit lower utility. One interpretation is gender discordance: OUD patients in Washington State are predominantly male, and patient-physician gender concordance is associated with better care experiences and outcomes across clinical settings (Greenwood, Carnahan and Huang, 2018). An alternative is practice-setting selection: female physicians treating OUD during the study period were more concentrated in primary care than in dedicated addiction medicine settings, potentially affecting treatment quality as perceived by the patient, though addiction medicine certification is separately controlled. Patients matched with more experienced providers or those specializing in addiction medicine exhibit higher utility, consistent with higher treatment quality or better alignment with patient needs. Provider characteristics also enter the search cost equation. More experienced providers are associated with higher patient search costs, contrary to the intuition that experienced providers might direct patients to better-stocked pharmacies. I therefore interpret the provider-experience estimates cautiously: the positive coefficient in both δ and μ indicates that provider experience is correlated with higher perceived treatment value and higher effective search costs, rather than isolating a clean provider-experience mechanism. One possibility is residual patient selection, with experienced providers treating clinically complex patients whose unobserved characteristics raise search costs even after conditioning on observed health history.

Health History: Patients whose OUD diagnosis appears earlier in the coding hierarchy (i.e., more likely to be the primary diagnosis) exhibit higher utility and face lower search costs, consistent with stronger treatment engagement among those most actively diagnosed. The estimates also reveal negative selection into treatment: patients with more severe health histories (higher

on urban ZIP code distributions.

Table 1— Estimated Coefficients

Preference Coefficients in δ			Search Coefficients in μ		
Variable	Estimate	SE	Variable	Estimate	SE
Out-of-Pocket Price	-0.005	2.7e-7	Distance from Residence	0.044	6.5e-5
×Medicaid	-0.408	0.002	×Urban	0.107	8.6e-5
×Medicare	-0.006	5.0e-7	Distance from Default	0.039	4.8e-5
			×Urban	0.039	6.4e-5
			Urban	0.036	0.027
Provider Characteristics					
Female Provider	-0.118	0.002	Female Provider	0.082	0.002
log(Experience)	0.043	4.0e-5	log(Experience)	0.045	4.0e-5
Addiction Medicine Specialist	0.025	0.004	Addiction Medicine Specialist	-0.006	0.004
× log(Past Opioid Consumption)	-0.033	2.6e-4	× log(Past Opioid Consumption)	-0.055	3.3e-4
× log(Past Mental Illness)	-0.002	0.019	× log(Past Mental Illness)	-0.004	0.022
× log(Past Substance Use Disorder)	0.000	0.012	× log(Past Substance Use Disorder)	-0.005	0.012
Health History					
OUD First Occurrence Position	-1.376	8.8e-4	OUD First Occurrence Position	0.469	0.003
# of All Conditions	0.026	1.2e-4	# of All Conditions	-0.061	1.2e-4
log(Past Opioid Consumption)	-0.314	5.4e-5	log(Past Opioid Consumption)	-0.242	7.4e-5
log(Past Mental Illness)	-0.035	0.004	log(Past Mental Illness)	-0.025	0.005
log(Past Substance Use Disorder)	-0.038	0.002	log(Past Substance Use Disorder)	-0.019	0.003
Demographics					
Black	-0.034	0.008	Black	0.026	0.007
Age	0.160	1.0e-4	Age	0.014	1.3e-4
Age Squared	-0.002	1.4e-8	Age Squared	-0.000	1.9e-8
Gender	-0.040	0.002	Gender	-0.045	0.002
Medicaid	-0.103	0.006	Medicaid	0.050	0.002
Medicare	-0.059	0.007	Medicare	0.032	0.008
Above High School	0.009	0.186	Above High School	0.026	0.328
log(Income)	0.126	0.010	log(Income)	0.313	0.008
Unemployment	0.005	0.878	Unemployment	-0.003	1.580
Poverty	0.013	0.168	Poverty	-0.005	0.160
Pharmacy Attributes					
Same Affiliation as Default	0.094	0.030	Same Affiliation as Default	-0.086	0.034
CVS	-0.041	0.056	CVS	0.035	0.060
Walgreens	-0.122	0.032	Walgreens	0.099	0.035
Rite Aid	0.169	0.029	Rite Aid	-0.151	0.031
Walmart	0.040	0.032	Walmart	-0.030	0.038
Albertson's	-0.030	0.031	Albertson's	0.044	0.033
Fred Meyer	-0.027	0.032	Fred Meyer	0.019	0.036
Bartell	-0.005	0.033	Bartell	0.020	0.036
Regional Chain	-0.010	0.037	Regional Chain	0.009	0.040
Constant	0.021	0.998	Constant	0.019	0.912

Note: Coefficient estimates obtained via maximum likelihood. Columns 1–3 display preference parameters; columns 4–6 report search parameters. Standard errors are heteroskedasticity-robust (Huber–White sandwich estimator).

past opioid use, mental illness, and substance use disorder) derive lower utility from buprenorphine. Those with the greatest clinical need thus appear least likely to value treatment, a pattern that raises concerns for both equity and effectiveness. The relationship reverses in the search cost equation: sicker patients face lower search costs. A natural explanation is that patients with multiple comorbidities already fill prescriptions for other conditions and are familiar with local pharmacy availability, reducing marginal search costs for buprenorphine. Patients with more diagnosed conditions show the same pattern, consistent with the same mechanism.

Demographics: Black patients face a dual burden: lower utility from the pharmacy visit and higher search costs. Lower utility may reflect stigma or negative experiences at the point of dispensing. Higher search costs are consistent with the geographic maldistribution of buprenorphine-stocking pharmacies: pharmacies serving predominantly Black neighborhoods stock buprenorphine at substantially lower rates (Weiner et al., 2023), requiring Black patients to search farther before finding a willing pharmacy. Together, the estimates imply that Black patients are less likely to fill a first prescription both because the dispensing experience is less valuable and because they face a longer effective search.

Age has an inverted-U effect on treatment preference, with utility peaking at approximately 40 years, consistent with the demographics of the opioid epidemic; search costs increase nearly linearly with age, consistent with reduced mobility. Female patients derive lower utility and face lower search costs; the lower search cost likely reflects greater familiarity with the healthcare system from navigating it for other health needs. Medicaid and Medicare recipients derive lower utility and face higher search costs, consistent with lower incomes and reduced access to transportation. Higher income predicts both higher utility and substantially higher search costs; the asymmetry suggests the income gradient in search costs captures opportunity cost of time beyond what income contributes to preferences. Education, unemployment, and poverty are measured at the ZIP-code level and are not statistically significant, likely reflecting the coarseness of the geographic proxy.

Pharmacy Attributes: Chain-brand effects are largely absorbed by the same-affiliation indicator: when a patient's default pharmacy belongs to a given chain, all locations of that chain receive a value of 1, capturing the brand-specific component of the default relationship. Patients filling at an affiliated pharmacy derive higher utility and face lower search costs, consistent with established relationships and greater familiarity. Among individual chains, Walgreens is associated with lower utility and higher search costs, while Rite Aid shows the opposite. The Rite Aid pattern may reflect a chain-wide practice of stocking buprenorphine more consistently, making it a more reliable destination for patients who have already identified it as an option.

In sum, on the cost side, travel imposes an effective search cost of $0.62 per additional mile, high relative to the out-of-pocket price of the drug. On the preference side, patients with the most severe clinical histories derive the least utility from buprenorphine, consistent with negative selection into treatment. The demographic gradient is equally sharp: Black patients face lower utility and higher search costs simultaneously, Medicaid and Medicare recipients face lower utility and greater travel barriers, and higher-income patients face elevated search costs consistent with higher opportunity cost of time.

B. Patient Search Behavior

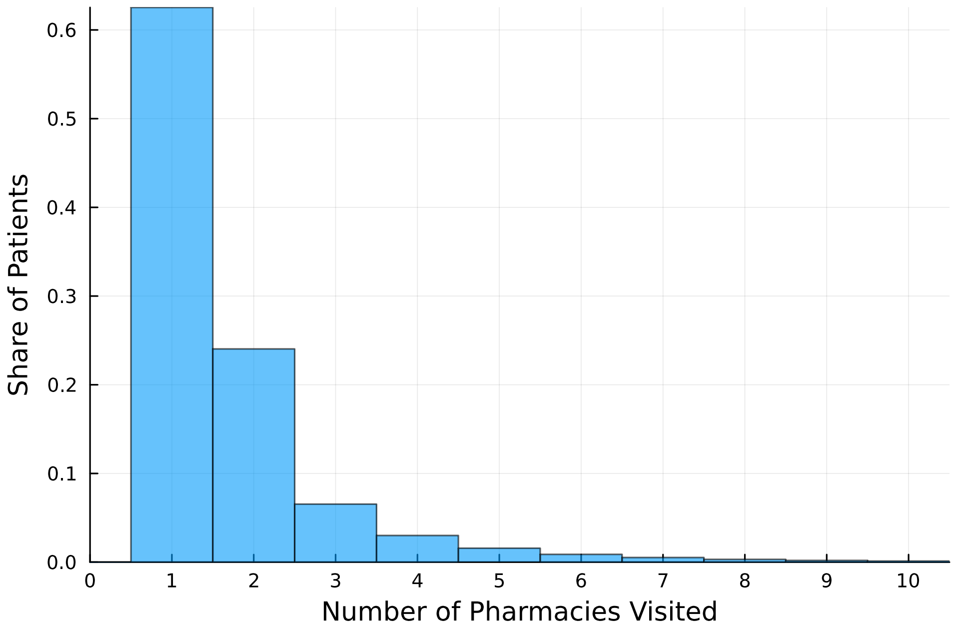
Although the model is not estimated from direct observations of search behavior, it can predict search patterns from the estimated parameters. For each patient, I simulate draws of utility and effective search costs from their estimated distributions, compute reservation values, and apply the Weitzman stopping rule to generate a predicted search sequence.

I focus on two dimensions: the number of pharmacies visited and the incidence of *suboptimal searches*, defined as visits to pharmacies without buprenorphine. Such visits reveal availability information but do not yield a fill.

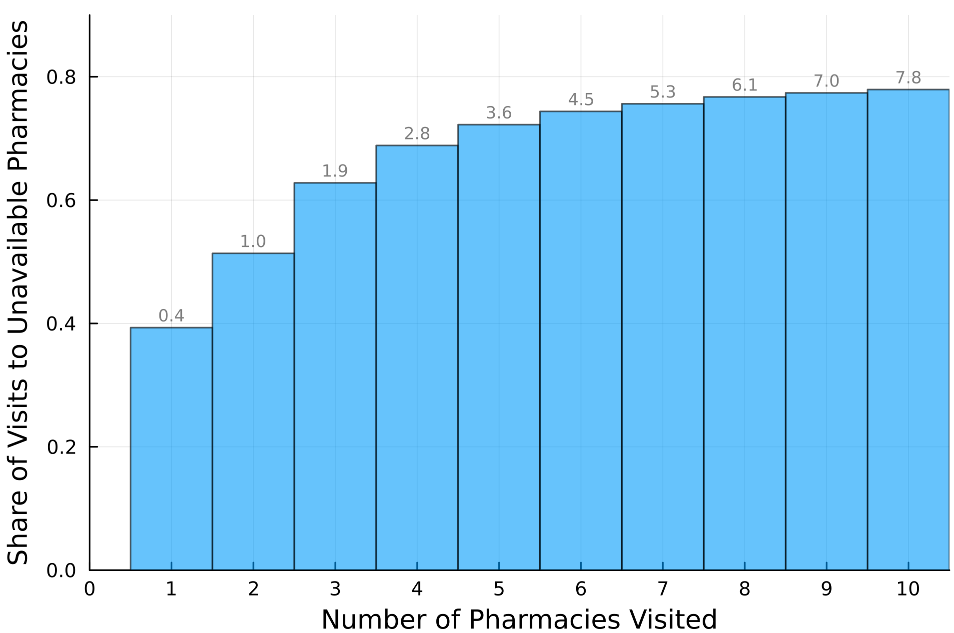
Figure 4a shows limited search: over 60% of patients search only once, which by assumption corresponds to visiting only their default pharmacy. The majority do not search further even when their default does not carry buprenorphine.[25] As a benchmark for the magnitude of these frictions, eliminating search costs entirely, equivalent to delivering buprenorphine to the patient's door, would raise the buprenorphine fill rate by 38 percentage points, from 45% to 83%, implying that effective search costs account for roughly 70% of treatment failures.[26]

[25] This pattern is consistent with other high-stakes search settings: Ambokar and Samaee (2019) find that 59% of U.S. mortgage borrowers make only one refinancing inquiry and about 80% make fewer than two.

[26] Home delivery is not a realistic policy lever for buprenorphine: as a controlled substance, it faces tight dispensing



(a) Number of Searches



(b) Number of Suboptimal Searches

Figure 4. Predicted Consumer Search Behaviors

Note: Predicted patient search behavior based on the estimated model. Panel (a) shows the distribution of the number of pharmacies visited. Panel (b) reports the average share of visits to pharmacies without buprenorphine available, conditional on total pharmacies visited. For each patient, 100 draws of utility and search costs are taken from the estimated distributions, and the Weitzman rule determines the search sequence and stopping point.

Figure 4b examines the extent to which patients visit pharmacies that do not have buprenorphine available. Patients are grouped by the total number of pharmacies visited, and for each group, the figure plots the average share of visits made to pharmacies without supply. For example, among patients who visited two pharmacies, one visit on average was to a pharmacy without buprenorphine, meaning that 50 % of their searches were unsuccessful. The share of unsuccessful visits increases with the total number of pharmacies visited. Among patients who visited only one pharmacy, 40 % encountered a location without supply. Among those who visited ten pharmacies, 78 % of visits were to locations that lacked buprenorphine. This pattern indicates that patients who search extensively are primarily seeking availability rather than engaging in search for match value, even though the model permits search for match value.

VII. Counterfactuals

Patients seeking buprenorphine face two reinforcing barriers: physical search costs and uncertainty about which pharmacies carry the drug. I evaluate counterfactual policies that reduce these barriers without imposing universal availability. Because beliefs ϕ_{ij} are not separately identified from physical search costs, I use a rational-beliefs calibration in which perceived availability equals observed local availability. The counterfactuals then reduce the uncertainty component of the effective search cost.[27]

The first set of counterfactuals isolates information provision. I compare a universal-availability benchmark with two more targeted policies: a centralized availability platform that prescribers can query, and informal networks in which prescribers track availability at a small set of nearby pharmacies. The second counterfactual evaluates e-prescribing. During the study period, controlled-substance prescriptions were handwritten and could not be transmitted in advance; e-prescribing instead routes the order directly to a chosen pharmacy, confirming availability before the patient travels. Because controlled-substance e-prescriptions cannot be transferred between pharmacies without a new prescription, the policy lowers uncertainty but raises switching costs

restrictions, and the troubled history of mail-order prescription opioids during the early phase of the epidemic makes broad expansion politically infeasible.

[27]Although ϕ_{ij} is not identified, its limiting cases are informative. If $\phi_{ij} = 1$ for every patient and pharmacy, patients already believe stocking is universal, so information provision has no effect on behavior. As ϕ_{ij} approaches zero, patients believe stocking is rare, so accurate information about specific pharmacies has a larger effect. Actual beliefs lie between these cases, so I report sensitivity to this assumption below.

when the first pharmacy is unavailable. The third counterfactual offsets that switching cost with small first-fill subsidies.

A. Information Provision Counterfactuals

I examine three policies that resolve uncertainty about pharmacy availability:

1) **Universal availability mandate.** All pharmacies are required to carry buprenorphine. The mandate eliminates uncertainty and expands the choice set to every pharmacy, providing an upper-bound benchmark for information- and access-side policies. Operational and financial constraints make this policy difficult to implement in practice.

2) **Real-time availability shared by prescribers.** The state maintains a pharmacy-level availability database that prescribers can query and share with patients at the point of care. The intervention eliminates uncertainty without expanding the choice set. Pharmacy discretion is preserved because availability is not advertised directly to patients.

3) **Informal prescriber–pharmacy networks.** Without a centralized system, prescribers form informal arrangements with a small number of nearby pharmacies, such as the five closest pharmacies that reliably carry buprenorphine. Patients may visit any pharmacy, but they know with certainty that buprenorphine is available at the network pharmacies.

All three policies shift beliefs ϕ_{ij} toward one, lowering the uncertainty component of the effective search cost. Under the rational-beliefs calibration, $\hat{\phi}_i$ denotes the share of pharmacies in patient i's choice set that carry buprenorphine. For example, if ten pharmacies are within reach and three carry it, $\hat{\phi}_i = 0.3$. Figure E1 reports the cross-patient distribution. Effective search costs scale inversely with $\hat{\phi}_i$, so patient i's search cost for pharmacy j is $\tilde{c}_{ij}/\hat{\phi}_i$.

For each patient, I draw 100 realizations from the adjusted search-cost distribution $F_{\tilde{c}}(\tilde{c}_{ij}/\hat{\phi}_i)$ and compute reservation values using Equation (5). I draw realized utility from a Type I Extreme Value distribution with location parameter $\hat{\delta}_{ij}$, and define latent utility w_{ij} as the minimum of the reservation value and realized utility. Patients then follow the Weitzman rule: they visit pharmacies in descending order of reservation value, stop once realized utility exceeds the next-best reservation value, and purchase from the pharmacy with the highest w_{ij}. Consumer surplus

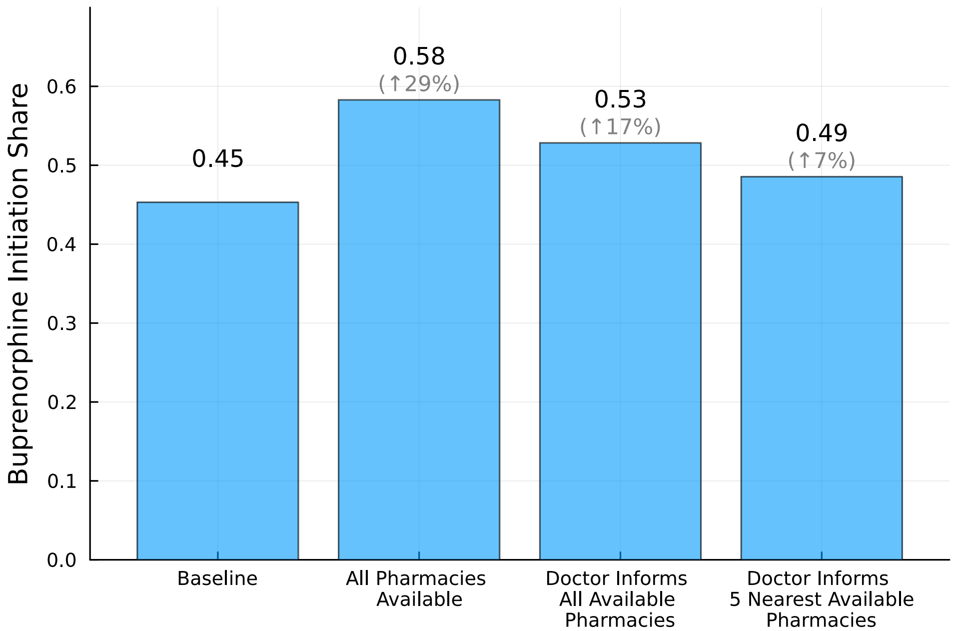


Figure 5. Buprenorphine Purchase Counterfactuals

Note: This figure reports the predicted population share purchasing buprenorphine under three counterfactual scenarios: (i) all pharmacies carry buprenorphine, (ii) prescribers have full knowledge of pharmacy availability, and (iii) prescribers know five nearby pharmacies that carry buprenorphine. Purchase shares are simulated by drawing search costs from $F_{\tilde{c}}(\tilde{c}_{ij}/\hat{\phi}_i)$ and computing w_{ij} for each patient. The baseline reflects actual purchase shares in the data. Numbers above each bar report the percentage increase relative to baseline.

equals realized utility at the chosen pharmacy minus cumulative search costs, scaled by the price coefficient to convert the measure to dollars.

Figure 5 reports predicted initiation under each scenario. Universal availability raises initiation from 45% to 58%, a 29% increase.[28] A more practical alternative is to give prescribers real-time availability information to share with patients at the point of care. This policy does not expand the choice set but eliminates uncertainty, raising initiation by 17%, about 58% of the universal-availability effect.

The two policies differ on one key margin: universal availability guarantees that the default pharmacy carries buprenorphine, whereas prescriber-informed availability does not. Yet information provision alone captures more than half of the universal-availability gain, implying that

[28]The gain is below the 83% ceiling that would obtain if both physical search costs and uncertainty were eliminated, equivalent to a home-delivery system in which patients face no search costs.

most missed initiations are not driven by the default pharmacy's inventory status. They are instead driven by uncertainty about which other pharmacies carry the drug. Once a prescriber identifies an alternative pharmacy known to carry buprenorphine, that pharmacy's effective search cost falls sharply under the rational-beliefs calibration. Patients whose default pharmacy does not carry the drug therefore become substantially more likely to initiate treatment.

Smaller-scale interventions also generate meaningful gains at lower implementation cost. If a provider maintains informal relationships with five nearby pharmacies known to carry buprenorphine and shares this information with patients, initiation rises by 7%. A handful of known alternatives lowers effective search costs even when uncertainty remains for the rest of the local market. The policy can be implemented at the provider level and requires no centralized infrastructure. Consumer surplus follows the same pattern (Figure E2).

A back-of-the-envelope calculation translates these initiation gains into health and fiscal terms. Each additional initiate avoids about 0.6 emergency room visits per year, a 20% reduction from a pre-treatment baseline of three visits per year (Figure B3); gains roughly one quality-adjusted life year (QALY) over their lifetime (Murphy and Polsky, 2016); and faces about half the mortality risk while in treatment (Larochelle et al., 2018).[29] Using Premier's disposition and cost figures, I set the average overdose-related emergency room hospital cost to $3,653.37 per visit.[30] Qian et al. (2024) simulate that, per 100 individuals entering office-based buprenorphine treatment, status-quo care averts about 13 overdoses, 2 of them fatal, over five years relative to no treatment.

The three counterfactuals imply different numbers of new initiates per 100 first-time OUD patients with buprenorphine prescription: universal availability adds 13, prescriber-informed availability adds 8, and informal networks add 3. For every 100 additional initiates, the policy averts about 60 emergency room visits per year, reduces annual hospital costs by $219,180, produces roughly 100 lifetime QALYs, and prevents about 13 overdoses, including 2 fatal overdoses, over five years. Even the smallest policy generates meaningful health and fiscal returns; the larger policies scale with the number of additional initiates.

[29] A QALY is one year in perfect health, the standard health-economics outcome for cost-effectiveness comparisons; one QALY can also reflect, for example, two years at 50% health-related quality of life.

[30] Computed as $0.47 \times 504 + (1 - 0.47) \times (11{,}731 + 20{,}500) \times 0.2 = 3{,}653.37$. Inputs come from Premier Inc. analysis summary [Accessed: 2026-04-23]: 47% treated and released, 53% treated and admitted, and average costs of $504 (treated and released), $11,731 (treated and admitted), and $20,500 (ICU).

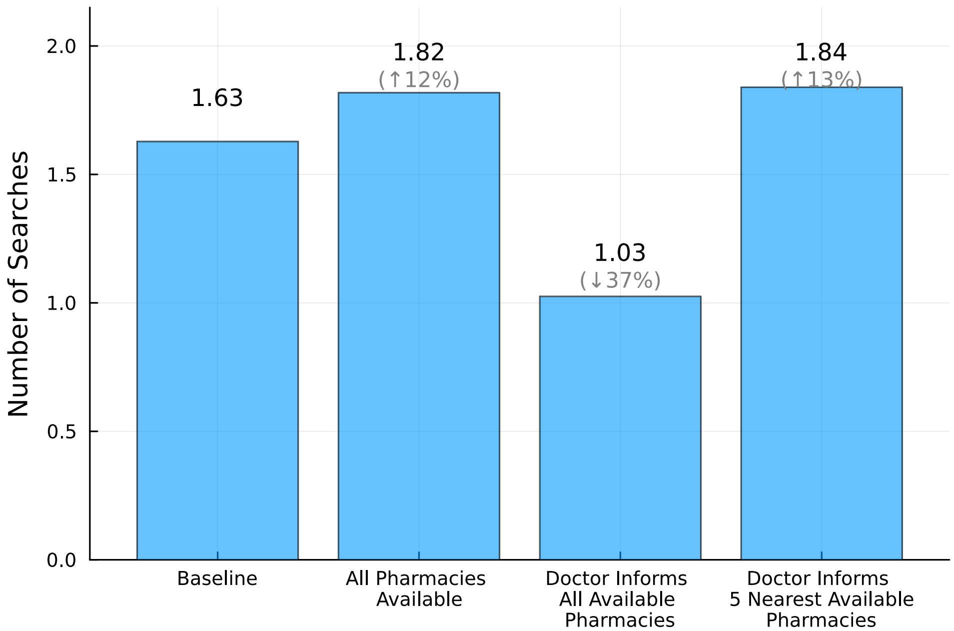


Figure 6. Number of Searches Counterfactuals

Note: Simulated analogously to Figure 5, showing the predicted number of pharmacies searched under each policy scenario. The baseline reflects actual search behavior in the data.

Figure 6 shows how search behavior responds. Universal availability raises searches by 12%: lower effective search costs induce some patients to look beyond the first available pharmacy, though most still stop after finding a pharmacy that carries the drug. Prescriber-informed availability reduces searches by 37% because patients no longer spend visits on pharmacies that do not carry buprenorphine, including their default pharmacy. The informal-networks policy raises searches by 13% through the same mechanism as universal availability, but only within the small set of known-availability pharmacies.

The rational-expectations assumption may be strong. If patients are naive and presume pharmacies always carry buprenorphine, observed behavior would reflect only physical search costs and information provision would appear ineffective. To assess robustness, I rescale perceived availability to $s\hat{\phi}_i$ for $s \in [0.1, 2]$, where $s < 1$ corresponds to pessimistic beliefs and $s > 1$ to optimistic beliefs, and recompute initiation under the prescriber-informed availability scenario. Figure E3 shows two patterns: the response is asymmetric, with pessimism raising initiation more than optimism reduces it, and information provision raises initiation modestly even under very optimistic beliefs.

As an additional check, I estimate ϕ_{ij} directly from each pharmacy's realized stocking history, treating patients as if they formed beliefs from perfect knowledge of past availability. The resulting counterfactuals closely match the rational-beliefs baseline.[31] Appendix F reports all counterfactual results under this alternative assumption.

B. Electronic Prescribing

By 2025, e-prescribing for controlled substances had become a central policy tool: federal rules require electronic prescribing for Schedule II–V controlled substances covered under Medicare Part D and Medicare Advantage prescription drug plans, and many states have adopted broader e-prescribing mandates.

E-prescribing also introduces a new barrier. For controlled substances, prescriptions generally cannot be transferred between pharmacies without the prescriber issuing a new one.[32] If the selected pharmacy is out of stock, the patient must contact the prescriber again, raising switching costs. This burden is especially consequential for buprenorphine because many pharmacies do not carry it.

I model e-prescribing as two opposing forces: lower uncertainty about pharmacy availability and higher switching costs when a prescription must be redirected. Because the burden of contacting a provider for a new prescription is unknown, I ask how much search costs would need to rise to offset the informational gain. Let s denote the multiplier on the search-cost location parameter μ, so $s = 2$ doubles costs. Uncertainty is captured by $\hat{\phi}_i$, and search costs are simulated from $F_{\tilde{c}}(s\tilde{c}_{ij}/\hat{\phi}_i)$.

Figure 7 shows how predicted initiation varies with s. At $s = 1$, predicted initiation matches the prescriber-informed availability scenario. As s increases, initiation falls because redirecting prescriptions becomes more costly. For e-prescribing to reduce initiation below the paper-based

[31] The "universal availability" counterfactual is highly sensitive to beliefs. Many pharmacies historically did not carry buprenorphine, so patients plausibly assign them very low availability. Under the policy, perceived availability at those locations rises from near zero to one, eliminating the uncertainty component of search and sharply lowering effective search costs. This sensitivity is specific to the universal-availability scenario; in the other counterfactuals, pharmacies with low prior availability remain unavailable and therefore do not enter the choice set, as they do not carry buprenorphine.

[32] This transfer restriction applies to controlled substances, not to most other medications.

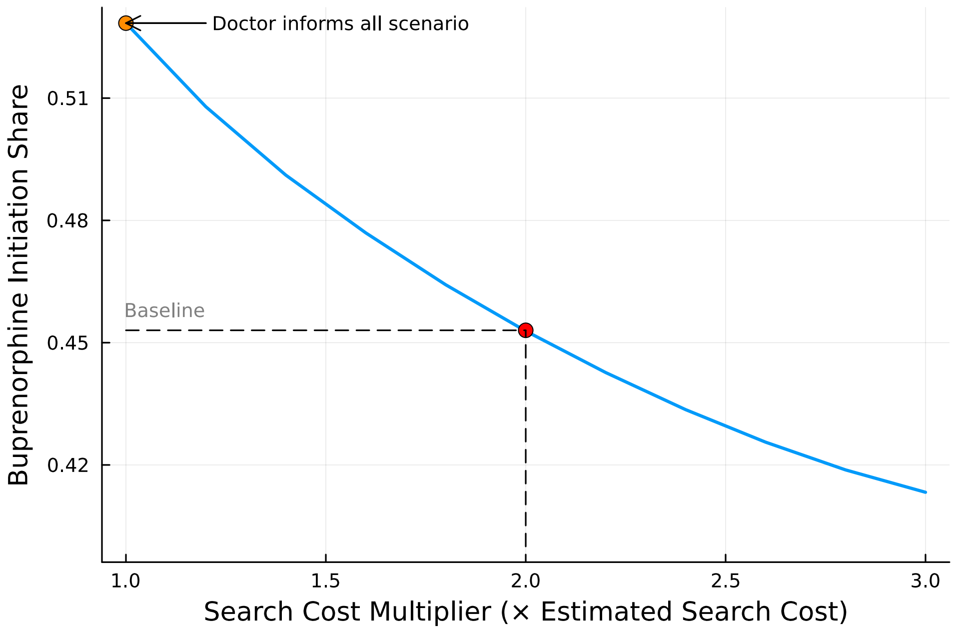


Figure 7. Trade-Offs Under E-Prescribing

Note: Predicted buprenorphine initiation under the e-prescribing counterfactual as the switching-cost multiplier s varies. At $s = 1$, the scenario matches the prescriber-informed availability counterfactual. As s increases, higher reauthorization costs offset the informational gain. The baseline reflects actual purchase shares in the data. Search costs are simulated from $F_{\tilde{c}}(s\tilde{c}_{ij}/\hat{\phi}_i)$; reservation values and realized utility are computed as in the main counterfactuals.

baseline, search costs would need to double. Contacting a prescriber for a new prescription is unlikely to be as burdensome as physically visiting an additional pharmacy. Under the estimated model, the e-prescribing mandate raises buprenorphine initiation on net unless switching costs double.

C. Monetary Incentives

Reducing informational frictions substantially raises initiation, but even eliminating uncertainty does not close the gap between OUD and other chronic conditions. One remaining barrier is the physical cost of search, especially travel: the estimates imply a cost of \$0.62 per additional mile, about 3% of the average out-of-pocket cost for buprenorphine. I use fill rates for levothyroxine, a widely available chronic medication that does not carry buprenorphine's stigma or stocking constraints, as a benchmark for unencumbered initiation in the same patient popu-

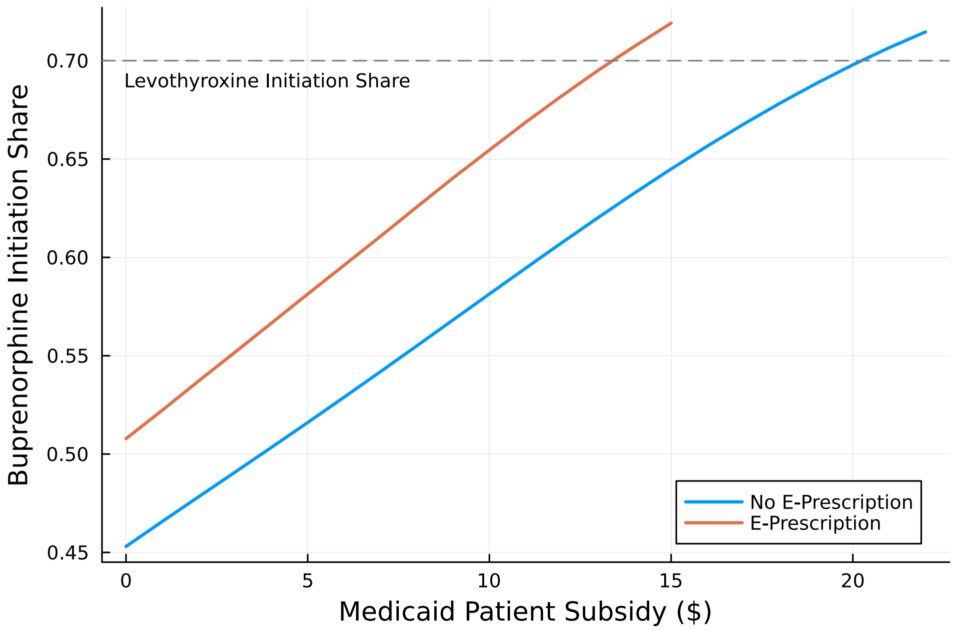


Figure 8. Medicaid Subsidy Counterfactual

Note: Predicted initiation share for Medicaid patients as the first-fill subsidy increases, with and without e-prescribing. Subsidies enter as direct utility transfers scaled by the Medicaid-specific price coefficient. Without e-prescribing, purchase probabilities follow Equation 12. With e-prescribing, purchase decisions follow the Weitzman rule, and search costs are simulated from $F_{\tilde{c}}(1.2\tilde{c}_{ij}/\hat{\phi}_i)$ to reflect reauthorization costs when the selected pharmacy is unavailable.

lation. I therefore ask how much direct monetary incentives could further increase initiation by offsetting these travel-related search costs.

Several states, including California and West Virginia, have piloted modest financial incentives to promote treatment adherence.[33] These programs typically provide gift cards of $10–$20 at treatment milestones, subject to a federal cap of $599 per patient per year. I evaluate whether similar subsidies could raise initiation at the first step of treatment. Because the first fill substantially increases the likelihood of continued adherence, targeting this margin may be cost-effective.

I focus on Medicaid patients, who represent roughly 40% of all individuals with OUD, are

[33]California's Recovery Incentives Program, implemented under the CalAIM Section 1115 Demonstration, provides contingency-management incentives in the form of gift cards up to $599 annually for patients with substance use disorder (California's Contingency Management Benefit [Accessed: 2025-08-12]). West Virginia's Section 1115 SUD Waiver similarly authorizes gift-card incentives for substance use disorder treatment under Medicaid Section 1115 Demonstration [Accessed: 2025-08-12].

the most price-sensitive group in the estimated model, and remain the least likely to initiate buprenorphine. For the no-e-prescribing scenario, I augment utility with a direct subsidy: I add \$x, scaled by the Medicaid-specific price coefficient, to the w_{ij} term in Equation (10), and recompute purchase probabilities using Equation (12). For the e-prescribing scenario, I incorporate the same subsidy into the Weitzman simulation and calibrate the additional switching burden as a 20% increase in the search-cost location parameter μ.

Figure 8 reports Medicaid initiation across subsidy levels, with and without e-prescribing. Without e-prescribing, a \$20.24 subsidy raises initiation to 70%.[34] With e-prescribing in place, a \$13.49 subsidy raises initiation to levels comparable to other chronic conditions. These amounts are within the range of existing gift-card programs, suggesting that first-fill incentives could be implemented using familiar policy tools. Pairing e-prescribing infrastructure with small first-fill incentives could therefore help close the last mile to treatment for Medicaid patients.

VIII. Conclusion

Buprenorphine remains underused at treatment initiation because the last mile is difficult to navigate. Even with a prescription in hand, patients face two search frictions: the physical cost of visiting pharmacies and uncertainty about which pharmacies stock the medication. I develop a sequential search framework that quantifies these frictions without observing search sequences. The identification comes from a single asymmetry: patients can search their default pharmacy at zero cost, so filling at another pharmacy reveals that the alternative's utility net of search cost exceeded the default's. These deviations from the default supply the variation needed to separately identify preferences and search costs.

The counterfactuals show that administratively simple tools can raise initiation to levels comparable to those for other chronic conditions. E-prescribing is now mandatory, so its informational benefits are already in place; pairing them with modest first-fill subsidies further reduces the physical search costs that deter initiation. Together, these levers can convert more prescriptions into treatment starts using tools already present in many prescribing and incentive programs.

[34] This result is notable because even under perfect naivete, where patients believe every pharmacy is available and thus face no information friction, a \$20 subsidy is sufficient.

Data Construction

A1. Patient Identification

I identify patients who initiated buprenorphine treatment for opioid use disorder (OUD). Ideally, initiation appears as an outpatient medical claim for OUD followed within 14 days by a buprenorphine pharmacy claim. Two complications arise: buprenorphine can be prescribed without an explicit OUD diagnosis, and an OUD diagnosis may lead to methadone or non-pharmacologic treatment rather than buprenorphine.

I address these issues in four steps. First, I exclude patients with any prior OUD-related care (ICD-9/10 codes in medical claims) or buprenorphine prescriptions (NDC codes in pharmacy claims) before April 1, 2014. I restrict to outpatient visits and to providers who prescribed buprenorphine during the same calendar year, retaining only new episodes of care plausibly associated with initiation.

Among these patients, I classify the first outpatient medical claim with an OUD diagnosis as a potential initiation event. If followed by a buprenorphine fill within 14 days, I classify the patient as matched.

For patients with a buprenorphine fill but no preceding qualifying medical claim, I search for the closest outpatient visit within 14 days of the fill date, prioritizing visits with the same prescriber recorded in the pharmacy claim and, among those, the visit closest in time.

Finally, for patients with a qualifying OUD visit but no timely pharmacy match, I check for sustained provider follow-up. If the patient returned to the same provider at least three times within 90 days, I infer a different treatment plan (e.g., methadone) and exclude these cases.

This process yields three mutually exclusive groups:

1) **Matched OUD initiators:** First OUD-related medical visit followed by a buprenorphine fill within 14 days (22% of first-time patients).

2) **Pharmacy-first initiators:** Buprenorphine fill matched to a proximate outpatient visit with the same prescriber, even without an explicit OUD diagnosis in the medical claim (22% of first-time patients).

3) **Unmatched medical visits:** OUD-related outpatient visits with no buprenorphine initiation and no sustained provider follow-up suggestive of alternative treatment (56% of first-time patients).

A2. Inventory Construction

I construct pharmacy-level buprenorphine inventory under two simplifying assumptions. First, I aggregate across formulations and strengths, treating all products (monotherapy, buprenorphine/naloxone combinations, and doses of 2mg, 8mg, and 12mg) as interchangeable in total milligrams. Clinical substitution across formulations requires medical judgment and may not always be feasible; pharmacies may also be unable to split high-dose tablets to fulfill low-dose prescriptions. This aggregation likely overstates availability.

Second, shipment data from the Automated Reports and Consolidated Ordering System (ARCOS) and sales data from the Washington State All-Payer Claims Database (WA-APCD) are not directly comparable. ARCOS captures all shipments to pharmacies; WA-APCD covers only insured Washington residents, excluding cash payments and out-of-state patients. WA-APCD sales account for roughly 40% of total shipments on average. I evaluate stock on the relevant date: the pharmacy claim date when available, otherwise the medical claim date. For each pharmacy in the patient's choice set, I compute the difference between cumulative shipments over the prior seven days (scaled by 0.403 to reflect the WA-APCD coverage share) and cumulative WA-APCD sales over the same window. A pharmacy has sufficient stock if this estimate exceeds the patient's inferred need. When a pharmacy claim is observed, I use the actual strength and duration of that claim; otherwise, I impute demand as a 14-day supply of 8mg buprenorphine/naloxone, the most common treatment pattern in the data.

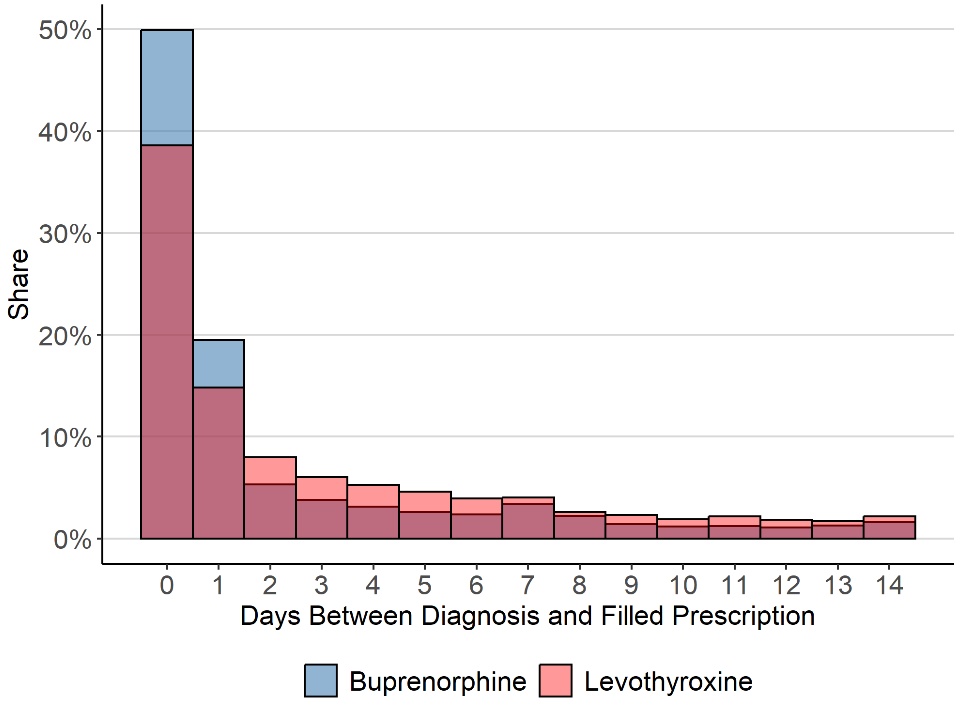


Figure B1. Number of Days Between Doctor Visits and Prescription Filled

Note: This figure displays the distribution of days between the first OUD-related medical visit and the first buprenorphine fill, and between the first hypothyroidism-related visit and the first levothyroxine fill. Medical conditions are sourced from WA-APCD.

Additional Stylized Facts

B1. Time to Fill a Prescription After Diagnosis

Timing matters for buprenorphine in a way it does not for most medications. Patients must begin treatment at the onset of withdrawal, and even brief delays raise the likelihood of relapse, so search frictions carry direct clinical consequences.

I compare time-to-fill for buprenorphine and levothyroxine among first-time users of each drug, measured as days between the first diagnosis-related visit and the first fill.

Figure B1 shows that most patients in both groups fill within a few days. Among OUD patients, nearly 70% obtain buprenorphine within one day of diagnosis; 55% of hypothyroidism patients fill levothyroxine within the same window. The faster initiation for buprenorphine reflects clinical urgency: when the default pharmacy does not stock it, even a brief search delay risks relapse. To rule out patient-population differences, I restrict both samples to individuals with at least four prior opioid prescriptions. Results are similar when further conditioning on patients who eventually fill buprenorphine, though the sample is smaller.

B2. Search Patterns Following Unavailability at the Default Pharmacy

I document patterns consistent with additional search when buprenorphine is unavailable at the default pharmacy. The search process is not directly observed, but subsequent purchase behavior reveals the search path.

I focus on patients who filled a prescription after encountering unavailability at their default pharmacy. For each such patient, I compute two distances for the pharmacy where the fill occurred: distance to the patient's residential ZIP code and distance to the default pharmacy. If patients check the default first, the observed fill reflects continued search. Proximity to home captures baseline convenience; proximity to the default pharmacy captures search initiated from that point.

Figure B2 shows that both distances skew toward shorter values: most patients fill close to home and close to the default pharmacy. The distribution is more concentrated near zero for distance to the default pharmacy, indicating that the default's location shapes the subsequent search path. These patterns support a search-based model in which distance plays a central role in pharmacy choice.

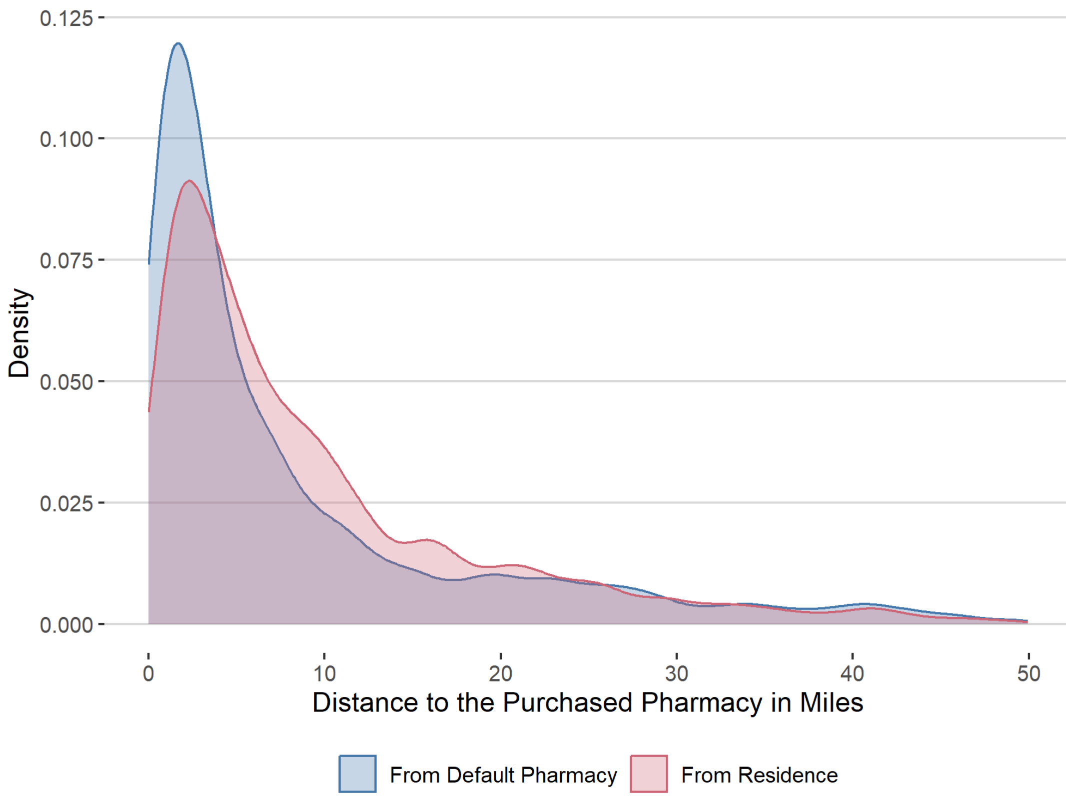


Figure B2. Distance to Default and Residential ZIP Code for Purchased Pharmacy

Note: This figure shows the distribution of distances from the purchased pharmacy to the default pharmacy and the patient's residential ZIP code, conditional on the default pharmacy being unavailable. The sample includes patients who ultimately filled a buprenorphine prescription elsewhere.

B3. Health Outcomes

The paper focuses on increasing buprenorphine initiation, but a natural question is whether initiation improves health outcomes. Existing evidence is mixed, relying largely on observational studies and clinical trials with limited external validity. This paper's design offers a more credible identification strategy. A direct comparison of patients who do and do not initiate is subject to selection bias: those who initiate are likely more health-conscious and may fare better even without medication. To address this, I instrument for treatment using default pharmacy availability. Patients are more likely to initiate when buprenorphine is available at their default pharmacy, and availability should affect health outcomes only through initiation. Combined with a difference-in-differences design that absorbs fixed unobservables, this yields a more credible estimate of the treatment effect.

The sample consists of first-time OUD patients with a buprenorphine prescription. The treatment group includes patients who initiate buprenorphine; the control group includes those who never fill the prescription.[35] The DiD specification is:

$$y_{it} = \beta \text{Buprenorphine Treatment}_{it} + \alpha_i + \gamma_t + \text{Doctor}_i + \text{ZIP}_i + \epsilon_{it} \tag{B1}$$

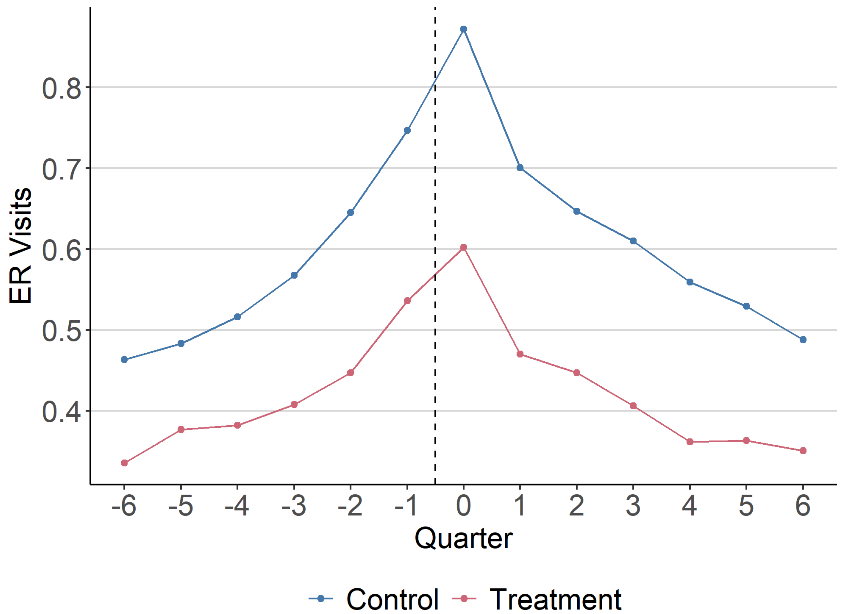
where y_{it} is the number of emergency room visits per quarter, Buprenorphine Treatment$_{it}$ is a binary indicator for treatment, α_i and γ_t are individual and time fixed effects, and Doctor$_i$ and ZIP$_i$ are doctor and ZIP code fixed effects.

The first stage regresses treatment on default pharmacy availability:

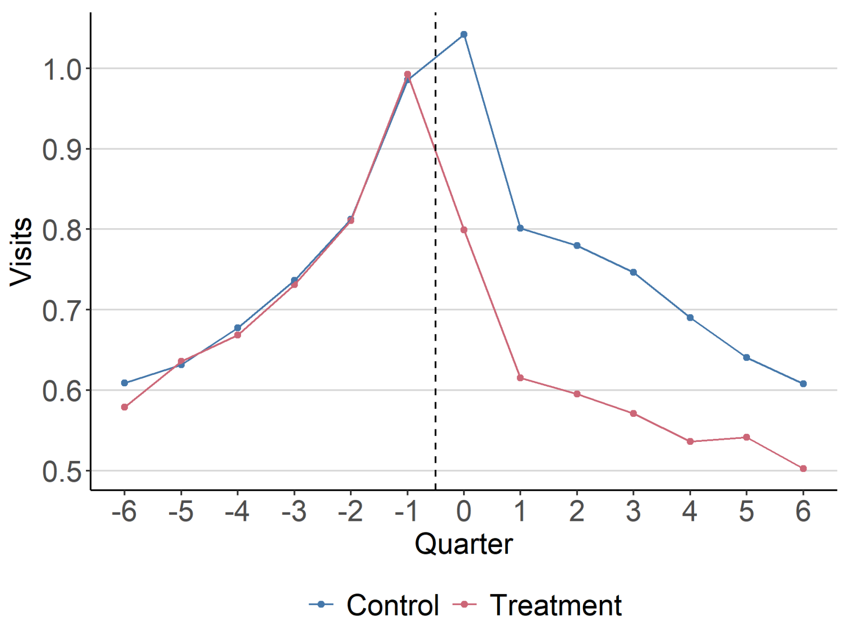
$$\text{Buprenorphine Treatment}_{it} = \pi \text{Default Availability}_{it} + \tilde{\alpha}_i + \tilde{\gamma}_t + \widetilde{\text{Doctor}}_i + \widetilde{\text{ZIP}}_i + \eta_{it} \tag{B2}$$

Figure B3 presents ER visit trends for treatment and control groups. Three patterns emerge from the raw trends (Figure B3a). First, patients do not enter OUD care immediately after an

[35]Later-treated patients are excluded from the control group to prevent bias from differential health trends.



(a) Raw Trends



(b) Partial Trends (60th percentile to 90th percentile)

Figure B3. Emergency Room Visits Per Quarter Trends by Treatment Groups

Note: This figure plots trends in emergency room visits for patients who received a buprenorphine prescription and either initiated treatment (treatment group) or did not (control group). ER visits are identified in the WA-APCD medical claims using place-of-service codes, with multiple claims on the same date counted as a single visit. Panel (a) shows raw trends, while Panel (b) restricts to patients between the 60th and 90th percentiles of pre-treatment ER visit frequency.

ER visit; there is a buildup period before treatment begins. Second, ER visits decline sharply after diagnosis regardless of whether the patient initiates buprenorphine. Third, patients who eventually receive buprenorphine already have lower ER visit rates before treatment begins. This last pattern matters for identification: using the full sample directly risks underestimating the treatment effect, since reductions are more pronounced among those starting from a high baseline.[36]

To ensure comparability, I stratify patients into quintiles of pre-treatment ER visit frequency. Figure B3b shows that restricting to the 60th–90th percentiles produces treatment and control groups that are more similar at baseline; the quintile stratification extends this logic to the full sample.

Figure B4 presents DiD and DiD-IV estimates by quintile. Across all quintiles, most estimates show statistically significant reductions in ER visits, in many cases exceeding 20% of the pre-treatment mean.

The DiD-IV estimates are consistently larger in magnitude, though not statistically different from the DiD estimates. The gap is largest in the bottom two quintiles, where DiD estimates are near zero but DiD-IV identifies substantial reductions. This reflects the LATE interpretation: DiD-IV scales the reduced-form effect by the first-stage compliance rate, isolating the effect for compliers (patients whose initiation depends on default pharmacy availability).

Always-takers, who would obtain buprenorphine regardless of pharmacy availability, are likely more motivated to seek treatment, so their average treatment effect is smaller. Standard DiD mixes compliers and always-takers, attenuating the estimate relative to DiD-IV.

[36] An analogy is weight-loss treatments. If drug A is given to severely overweight patients and drug B to moderately overweight patients, greater weight loss among drug A patients need not mean it is more effective; it may simply reflect differences in baseline weight.

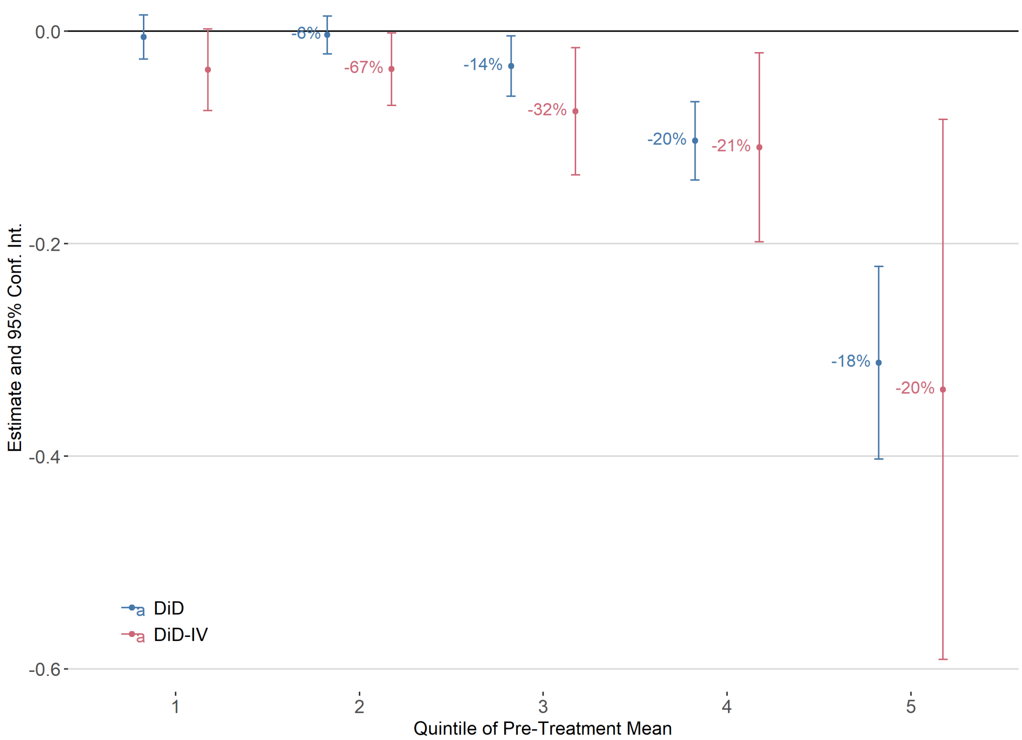


Figure B4. Taking Buprenorphine Effects on Emergency Room Visits

Note: This figure plots estimated effects of buprenorphine initiation on emergency room visits across quintiles of pre-treatment ER utilization. DiD estimates come from Equation B1; DiD-IV estimates use default pharmacy availability as an instrument for treatment. Ninety-five percent confidence intervals are clustered at the patient level.

Proof

The equivalence result applies to the information structure in the main model: consumers learn pharmacy availability only after visiting a pharmacy. If availability were observed before search, unavailable pharmacies would be removed before the consumer forms the search order, and the proof would reduce to the static-availability case. The case considered here is the empirically relevant one because buprenorphine availability is not verified before the patient travels during the study period.

Theorem 1 (Eventual Purchase): *Let* $w_j \equiv \min\{\mathbf{1}\{j \in A\}u_j + \big(1 - \mathbf{1}\{j \in A\}\big)(u_0 - \iota), r_j\}$ *for each* j*, where* $\iota > 0$ *is an arbitrarily small penalty that strictly dominates an unavailable product by the outside option. Given* u_0*, the consumer purchases product* j *if and only if* $w_j > u_0$ *and* $w_j > w_k$ *for all* $k \neq j$*.*

Proof: *Sufficiency.* If $w_j > u_0$, product j is available (if unavailable, $w_j < u_0$). Given product j is available, she purchases product j because she is willing to visit at least one seller ($r_j > u_0$) and make a purchase ($u_j > u_0$). It remains to show that if $w_j > w_k$ when product j is available, then product k is not chosen.

- **Case 1**: Suppose product k is unavailable ($\mathbf{1}\{k \in A\} = 0 \implies w_k < u_0$), she has no incentive to visit seller k or purchase product k. I only need to consider when product k is available.

- **Case 2**: Suppose $r_k \leq u_k$, then $w_k = r_k$. The consumer visits seller k only after seller j because $r_k \geq w_j > r_k$. However, once she visits seller j, she has no incentive to visit seller k because $u_j > r_k$.

- **Case 3**: Suppose $r_k > u_k$, which implies that $w_k = u_k$. In this case, even if she visits seller j, she either recalls a previous product ($u_j > u_k$) or continues to search ($r_j > u_k$) and finds a better product ($u_j > u_k$).

Necessity. If $w_j < u_0$, then seller j is neither visited ($r_j < u_0$) nor is product j purchased ($\mathbf{1}\{j \in A\}u_j < u_0$), regardless of availability.

If $w_j < w_k$ for some $k \neq j$, and product j is unavailable, product j cannot be chosen ($w_j < u_0$). If j is available, $w_j < w_k$ implies product k is also available since $0 < w_j < w_k$. By the same logic as Cases 2 and 3, the consumer does not purchase product j. Q.E.D.

Internal Validation

Figure D1 compares predicted and observed buprenorphine purchases across three categories (purchase at the default pharmacy, purchase at a non-default pharmacy, and no purchase), conditional on default pharmacy availability. The largest discrepancy is at most 3%, arising when buprenorphine is unavailable at the default but purchased elsewhere. Because these categories are not moments targeted in estimation, the close alignment supports the model's use in counterfactuals.

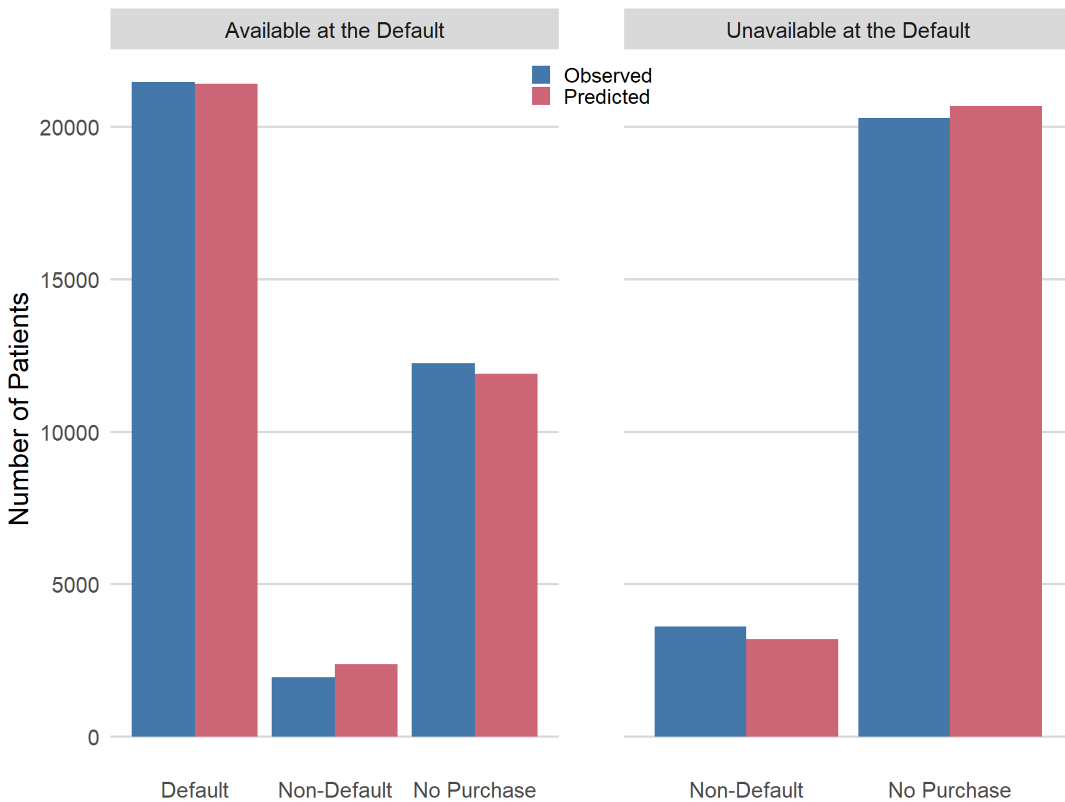


Figure D1. Internal Validation

Note: Observed and predicted number of patients purchasing at the default pharmacy, at a non-default pharmacy, or not at all, separately for cases when buprenorphine is available at the default pharmacy and when it is not. Predicted purchases aggregate individual purchase probabilities computed from the estimated coefficients in Table 1 via Equation 12.

Table E1— Summary Statistics

Variable	Mean	St. Dev.	Min	P25	Median	P75	Max
Individual Variables							
OUD First Occurrence Position	2.37	2.26	1.00	1.00	1.00	3.00	15.00
Past Opioid Consumption (MGE)	14,885.33	45,679.43	0.00	0.00	0.00	3,300.00	1,991,675.00
Past Mental Illness Diagnoses	0.70	2.46	0.00	0.00	0.00	0.00	92.00
Past Substance Use Disorder Diagnoses	0.93	3.41	0.00	0.00	0.00	1.00	92.00
# of All Conditions	4.05	2.25	1.00	3.00	4.00	5.00	25.00
Black	0.06	0.23	0.00	0.00	0.00	0.00	1.00
Age	41.85	14.57	0.00	30.00	39.00	53.00	90.00
Gender	0.52	0.50	0.00	0.00	1.00	1.00	1.00
Medicaid	0.55	0.50	0.00	0.00	1.00	1.00	1.00
Medicare	0.14	0.34	0.00	0.00	0.00	0.00	1.00
Above High School	0.90	0.06	0.00	0.88	0.91	0.94	1.00
Income	35,167.69	12,107.95	3,594.00	27,798.00	32,703.00	38,942.00	189,150.00
Unemployment	0.06	0.02	0.00	0.04	0.06	0.07	1.00
Poverty	0.13	0.06	0.00	0.08	0.12	0.16	1.00
Female Provider	0.39	0.49	0.00	0.00	0.00	1.00	1.00
Provider Experience	8.84	3.28	0.00	7.00	10.00	11.00	14.00
Addiction Medicine Specialist	0.19	0.39	0.00	0.00	0.00	0.00	1.00
Urban	0.92	0.28	0.00	1.00	1.00	1.00	1.00
Pharmacy Variables							
CVS	0.03	0.17	0.00	0.00	0.00	0.00	1.00
Walgreens	0.09	0.29	0.00	0.00	0.00	0.00	1.00
Rite Aid	0.09	0.29	0.00	0.00	0.00	0.00	1.00
Walmart	0.05	0.21	0.00	0.00	0.00	0.00	1.00
Albertsons	0.17	0.37	0.00	0.00	0.00	0.00	1.00
Fred Meyer	0.06	0.24	0.00	0.00	0.00	0.00	1.00
Bartell	0.05	0.22	0.00	0.00	0.00	0.00	1.00
Regional Chain	0.06	0.24	0.00	0.00	0.00	0.00	1.00
Individual-Pharmacy Variables							
Price	21.33	78.19	0.00	0.00	8.25	8.30	595.80
Distance from Default	15.52	9.43	0.00	8.42	14.52	21.16	50.00
Distance from Residence	15.07	8.26	0.02	8.90	14.78	20.48	50.00

Note: Summary statistics for the estimation sample of first-time OUD patients in Washington State (WA-APCD, 2014–2019). Individual characteristics are drawn from medical claims; pharmacy characteristics from ARCOS and WA-APCD pharmacy claims; individual-pharmacy variables are constructed at the patient-pharmacy-pair level.

Table E2— Placebo Tests Using Levothyroxine Refills

	Dependent Variable: Levothyroxine Refill Indicator	
	(1) 30 Days Before	(2) 30 Days After
Default pharmacy has buprenorphine available	-0.0014	0.0141
	(0.0278)	(0.0277)
Fixed effects		
Year	✓	✓
Pharmacy	✓	✓
Observations	2,939	2,939
R^2	0.351	0.377

Note: Each column reports a linear probability model. The dependent variable equals one if the patient refilled levothyroxine within 30 days before (column 1) or after (column 2) the index OUD visit. The regressor is an indicator equal to one if the patient's default pharmacy had buprenorphine available on the buprenorphine prescription date. Both columns include year and pharmacy fixed effects. Standard errors, in parentheses, are clustered at the individual member level. As a placebo test, default buprenorphine availability should not predict levothyroxine refill behavior if the main results do not simply reflect a general propensity to fill prescriptions. Sample: WA-APCD, 2014–2019.

Table E3— Buprenorphine Initiation and Default Pharmacy Availability

	Dependent Variable: Filled Buprenorphine			
	(1)	(2)	(3)	(4)
Default pharmacy has buprenorphine available	0.2841***	0.2933***	0.2806***	0.1423***
	(0.0035)	(0.0039)	(0.0039)	(0.0037)
Fixed effects				
Default pharmacy		✓	✓	✓
Month-year			✓	✓
ZIP				✓
Medical provider				✓
Observations	56,144	56,144	56,144	56,144
R^2	0.522	0.547	0.557	0.817

Note: Each column reports a linear probability model. The dependent variable equals one if the patient filled buprenorphine following the index OUD visit. The regressor is an indicator equal to one if the patient's default pharmacy had buprenorphine available on the prescription date. All specifications control for patient demographics, pharmacy chain affiliation, and provider characteristics. Column (1) adds no fixed effects. Column (2) adds default-pharmacy fixed effects, so identification comes from variation in availability across patients tied to the same default pharmacy. Column (3) adds month-year fixed effects. Column (4) further adds residential ZIP and medical-provider fixed effects. The estimates compare initiation rates when the same default pharmacy is stocked versus temporarily unstocked, holding fixed progressively richer sets of pharmacy, time, geographic, and provider factors. Standard errors, in parentheses, are clustered at the individual member level. ***, **, and * denote significance at the 1, 5, and 10 percent levels. Sample: WA-APCD, 2014–2019.

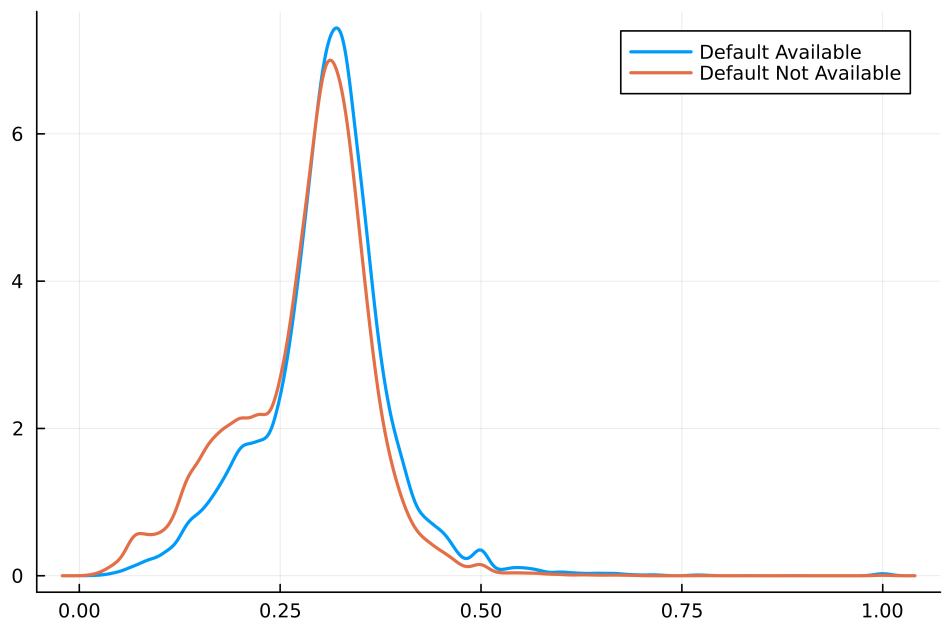


Figure E1. Distribution of Estimated Belief Parameter $\hat{\phi}$

Note: Distribution of the estimated belief parameter $\hat{\phi}_i$, patient i's prior probability that a randomly drawn pharmacy in their choice set carries buprenorphine. Distributions are shown separately for patients whose default pharmacy stocks buprenorphine and for those whose default does not. Beliefs are formed under rational expectations using neighborhood availability: $\hat{\phi}_i$ equals the share of nearby pharmacies that stock buprenorphine (e.g., if 3 of 10 local pharmacies stock it, $\hat{\phi}_i = 0.30$).

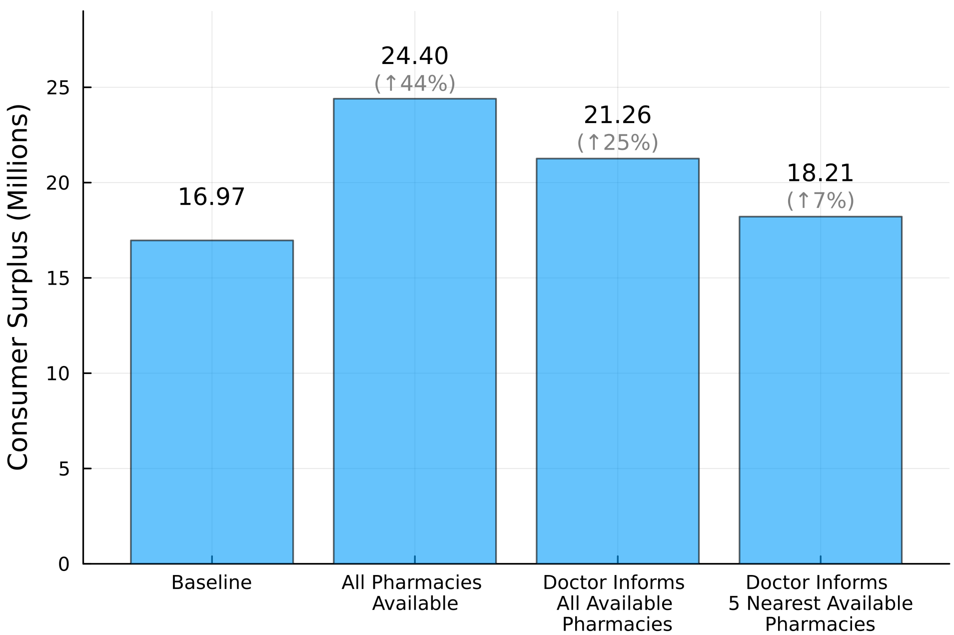


Figure E2. Consumer Surplus Counterfactuals

Note: Counterfactual consumer surplus across policies, simulated analogously to Figure 5. Consumer surplus equals realized utility minus cumulative search costs, scaled by the price coefficient to convert to dollars.

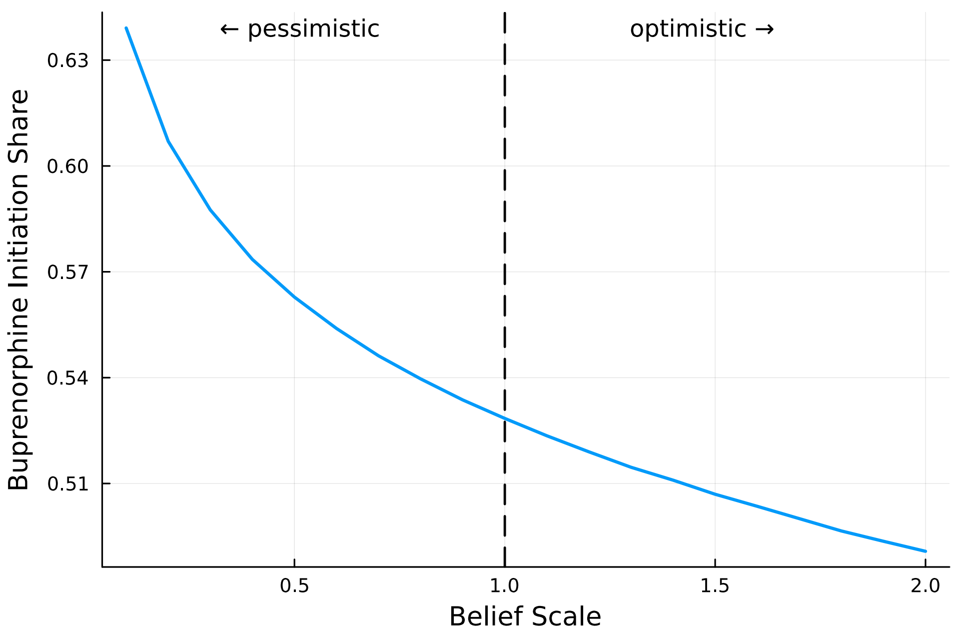


Figure E3. Pessimistic vs. Optimistic Beliefs

Note: The figure plots the predicted buprenorphine purchase share as the belief scale s varies from 0.1 (pessimistic) to 2 (optimistic) under the counterfactual scenario where prescribers inform patients about which pharmacies carry buprenorphine. The vertical line at $s = 1$ marks baseline beliefs.

Counterfactual Robustness

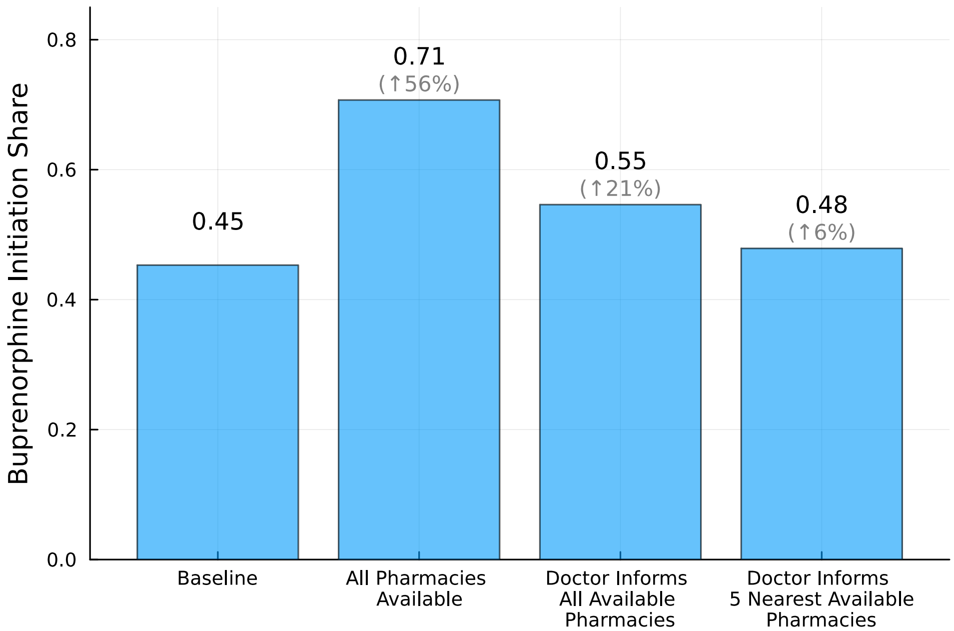


Figure F1. Buprenorphine Purchase Counterfactuals Under Correct Beliefs

Note: Robustness check for Figure 5: the same counterfactual exercise under the alternative assumption that each patient holds correct beliefs about the probability that each pharmacy in their choice set carries buprenorphine, based on knowledge of the pharmacy's past shipments. The baseline uses rational expectations based on local availability rates.

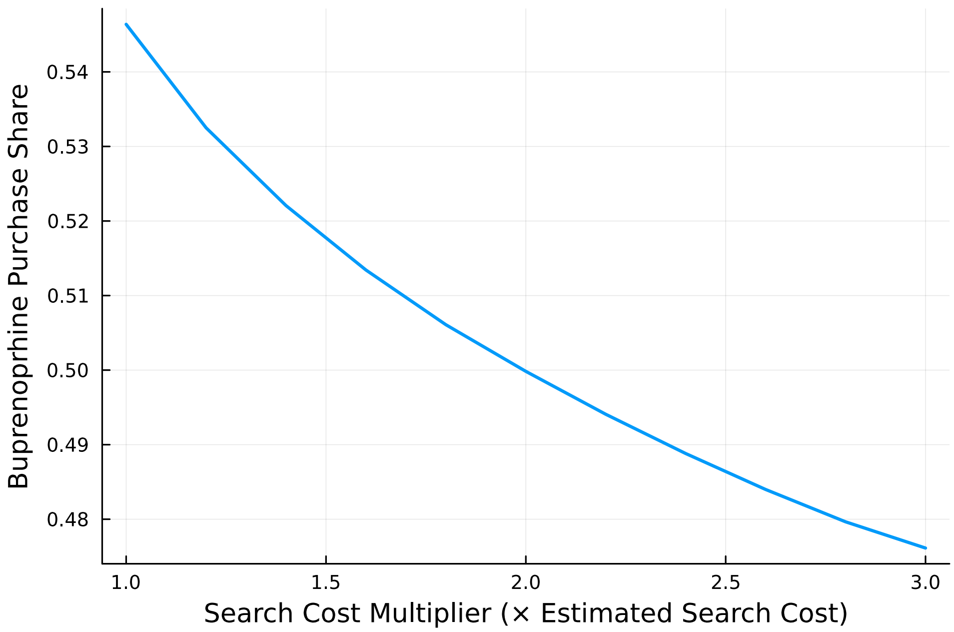


Figure F2. E-Prescription Counterfactuals Under Correct Beliefs

Note: Robustness check for Figure 7: the same counterfactual exercise under the alternative assumption that each patient holds correct beliefs about the probability that each pharmacy in their choice set carries buprenorphine, based on knowledge of the pharmacy's past shipments. The baseline uses rational expectations based on local availability rates.

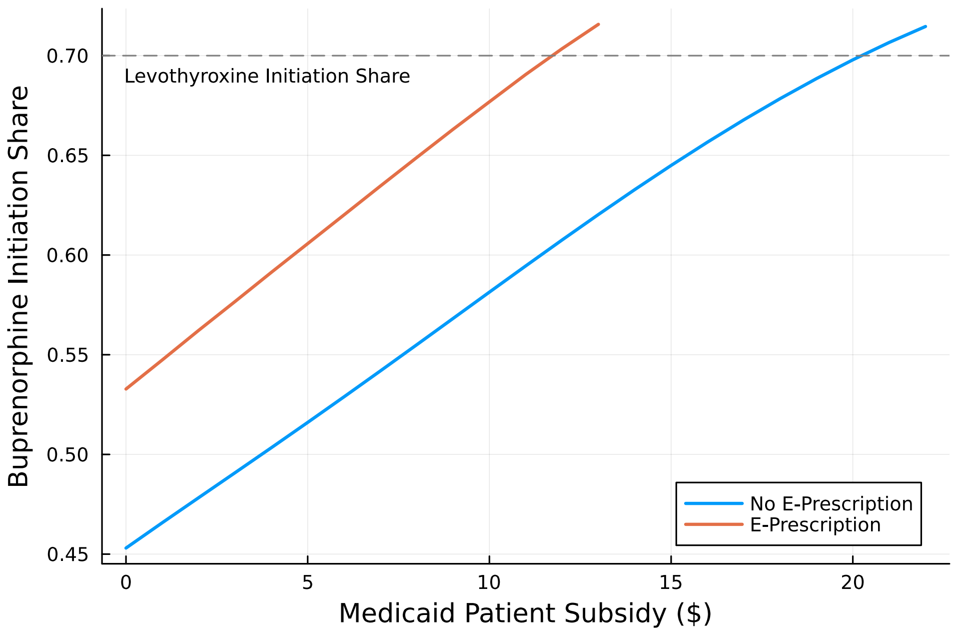


Figure F3. Subsidy Counterfactuals Under Correct Beliefs

Note: Robustness check for Figure 8: the same counterfactual exercise under the alternative assumption that each patient holds correct beliefs about the probability that each pharmacy in their choice set carries buprenorphine, based on knowledge of the pharmacy's past shipments. The baseline uses rational expectations based on local availability rates.

*

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